Testis, Penis, Urethra & Scrotum Rapid Revision

Campbell-Walsh-Wein · Chapters 78–87

For MCh Urology theory & viva · NEET-SS / competitive MCQs

How to use (total ≈ 5h 15min even at a slow, careful pace)

- Each chapter reads top-to-bottom in the order you would write an essay answer: definition → pathology → diagnosis → management → complications → follow-up. The same material serves theory and viva.
- Boxed **ALGORITHM** and shaded **KEY POINT** panels carry the reasoning examiners ask about ("why do we…?").
- **MCQ one-liners** end each chapter: re-read these the day before the paper.
- The last section puts the big treatment choices side by side for essay questions.

Conventions: **underlined bold** = number to memorise · MCQ = exam trivia only · → *Ch 80* = cross-reference · † = not in the source chapter notes (added because it is essential in practice). Abbreviations are spelled out at first use in each chapter.

Chapter	Topic	Slow-reader time
78	Anatomy of the Retroperitoneum	26 min
79	Neoplasms of the Testis	48 min
80	Surgery of Testicular Tumours	34 min
81	Minimally Invasive RPLND	17 min
82	Tumours of the Penis	48 min
84	Inguinal Node Dissection	19 min
83	Tumours of the Urethra	28 min
85	Benign Disorders of the Penis & Urethra	28 min
86	Urethral Stricture Disease	29 min
87	Surgery of the Scrotum & Seminal Vesicles	32 min
★	Putting it together: testis, penis and urethra decisions	6 min
	Total (22124 words, at 70 words/min)	**5h 15min**

78 Surgical, Radiographic and Endoscopic Anatomy of the Retroperitoneum

Reading time ≈ 26 min (slow pace)

1 · DEFINITION, BOUNDARIES AND ORGANS

- **Retroperitoneum** = space behind the peritoneum. Boundaries: **anterior** posterior peritoneal reflection; **posterior** abdominal wall; **cranial** diaphragm; **caudal** extraperitoneal pelvic structures. Whether psoas and quadratus lumborum are boundaries or contents is debated.

	Primarily retroperitoneal	**Secondarily retroperitoneal**
Development	Form and stay behind peritoneum	Begin in mesentery; peritoneum regresses and fuses to posterior wall
Organs	Adrenals, kidneys, ureters, aorta, inferior vena cava (IVC), anal canal	Ascending/descending colon, duodenum (**except D1**), pancreas (**except tail**)
Space	Perirenal	Anterior pararenal

> **KEY POINT** Secondarily retroperitoneal organs lie **in front of** the kidney, so colon, duodenum and pancreas are always in the surgeon's path to the kidney.

2 · POSTERIOR ABDOMINAL WALL AND SURFACE LANDMARKS

Landmark	**Left kidney**	**Right kidney**
Upper pole	11th rib	12th rib (right lower)
Hilum	**L1**	
Lower pole	L3–L4	

- **Flank layers**: external oblique (inferomedial, "hands in pockets") → internal oblique (superomedial) → transversus abdominis → transversalis fascia.
- **Psoas major** T12–L5 → lesser trochanter (hip flexion, with iliacus). **Psoas minor** inconstant; inserts at pelvic brim, not femur. **Quadratus lumborum** posterolateral to psoas; stabilises 12th rib.
- **Lower ribs**: fracture → suspect retroperitoneal injury. 12th rib is tethered to L1–L2 transverse processes by the **costovertebral (lumbocostal) ligament**; dividing it frees the rib for upper retroperitoneal exposure in posterior approaches.
- **Intercostal bundle**: costal groove of the rib **above**; vein–artery–nerve top-down — nerve lowest, least protected.
- **Supracostal** (above 12th) percutaneous access may cross the pleura → more complications than subcostal.
- **Lumbodorsal fascia**: posterior, middle and anterior layers (anterior continuous with psoas fascia) fuse at the **12th rib tip** → standard flank access point.
- **Dorsal lumbotomy**: vertical incision through lumbodorsal fascia lateral to erector spinae and quadratus — enters retroperitoneum **without cutting muscle**.

3 · RETROPERITONEAL FASCIAE AND SPACES

Three fetal mesenchymal strata → three fasciae → three spaces.

Stratum	**Fascia**	**Organs**	**Space**
Outer	Transversalis fascia	Body wall	Posterior pararenal (posterior/lateral walls)

Stratum	Fascia	Organs	Space
Intermediate	Renal (Gerota's) fascia, anterior + posterior laminae	Genitourinary	Perirenal
Inner	Fusion fascia with dorsal peritoneum	Gastrointestinal	Anterior pararenal

- **Posterior pararenal space**: anterior = posterior lamina of Gerota; posterior/lateral = transversalis fascia; medial = their fusion — **must be incised to reach the hilum** in retroperitoneoscopic dissection.
- **Renal fascia**: anterior lamina (Toldt, prerenal; thinner) and posterior lamina (Zuckerkandl, retrorenal; thicker, seen more on imaging). MCQ Gerota described only the anterior, yet his name covers both. Laminae merge laterally as the **lateroconal fascia** (separates anterior and posterior pararenal spaces).
- **Perirenal space**: adrenal, kidney, ureter, perirenal fat, renal pedicle, gonadal vessels. Perirenal fat is fine, light yellow; pararenal fat coarse, yellow-orange — the colour change guides the plane during colon mobilisation.
- Perirenal cone is **open inferiorly** into the extraperitoneal pelvis and may communicate across the midline below the hilum → urine (urinoma), blood (haematoma, ruptured aneurysm) and pus (perinephric abscess) are contained laterally but **track caudally**.
- **Anterior pararenal space**: between anterior renal fascia and posterior parietal peritoneum; contains colon, pancreas, D2–D3. Entered medially from the **white line of Toldt** (lateral border of colonic mesentery fusion with posterior peritoneum).

4 · GASTROINTESTINAL RELATIONS (OPERATIVE HAZARDS)

- **Right**: pancreatic head anterior to IVC; D2 on the right hilum → **Kocher manoeuvre**. D3 crosses between SMA (front) and aorta.
- **Left**: pancreatic tail, spleen, stomach near upper pole; splenic vessels along upper pancreatic border — at risk when lifting the pancreas off renal fascia.
- Divide hepatocolic/splenocolic ligaments sharply; avoid over-retraction (liver/spleen injury).
- **Mesentery** (now an organ): close small mesenteric defects after bowel isolation — small defects carry higher internal-hernia risk.

5 · GREAT VESSELS

Aorta and branches

Great vessel compartment runs T12 to L4/5 between the perirenal spaces. Aorta enters at **T12**, left of IVC; bifurcates at **L4**.

Branch	Type	Level	Supplies
Inferior phrenic	Paired lateral	First, below hiatus	Diaphragm; superior adrenal
Celiac trunk	Anterior	T12	Foregut
Middle adrenal	Paired lateral	At SMA level	Adrenal
Superior mesenteric artery (SMA)	Anterior	L1–L2	Midgut
Renal	Paired lateral	L1–L2, just below SMA	Kidney; inferior adrenal
Gonadal	Paired **anterolateral**	Below renals	Testis/ovary
Lumbar	Paired posterior	Four pairs, L1–L4	Wall, spine, cord
Inferior mesenteric artery (IMA)	Anterior	L3–L4	Hindgut

- Pattern: anterior unpaired → gut; paired lateral → adrenals, kidneys, gonads (which share lymphatic drainage); posterior → wall.
- **Adrenal**: superior (inferior phrenic, constant), middle (aorta), inferior (renal) — latter two variable.
- **Renal arteries**: supernumerary in ≥ ¼ → *Ch 84*.
- **Testicular artery** runs anterior to psoas, genitofemoral nerve, ureter (and IVC on right) to the deep ring.
- **Ureteric supply**: proximal renal; middle aorta, common iliac, gonadal; distal internal iliac (vesical). Abdominal ureter supplied **medially**, pelvic ureter **laterally** → mobilise accordingly.

Vessel	Safe to sacrifice because…
IMA (colon)	Marginal artery of **Drummond** (middle colic ↔ left colic)
IMA (rectum)	Superior ↔ middle/inferior rectal (internal iliac)
SMA	**Not safe** — severe bowel ischaemia (risk in left-sided surgery)

IVC and veins

- IVC forms below and right of the aortic bifurcation, ascends right of aorta, passes the central tendon at **T8**. Veins vary more than arteries.
- **Left renal vein**: long, crosses anterior to aorta below SMA; receives gonadal (caudal, right-angle entry), adrenal (superior; ± inferior phrenic) and ≥ 1 **lumbar vein** (posterior — avulsion risk). **Right renal vein**: short, usually no tributaries; duplicated in one-sixth. Renal veins anterior to arteries, enter IVC at L1.
- **Right gonadal vein** → anterolateral IVC (right renal vein in up to 10%); **right adrenal vein** short, straight into posterolateral IVC.
- **Left varicocele** predominance: longer vein + perpendicular entry. **New right varicocele** → image for retroperitoneal mass (e.g. renal cell carcinoma with venous thrombus).
- **Lumbar veins** drain via ascending lumbar veins (behind psoas) → azygos/hemiazygos.
- Right ureter: dissect **lateral** to the gonadal vein; medial plane risks IVC injury and gonadal vein avulsion.
- **Inferior mesenteric vein** lies in the left mesocolon; joins superior mesenteric vein (SMV) in 40%. **Reoperative trap**: mistaken for gonadal vein.

Variants (essential)

Variant	Anatomy	Significance
† Retroaortic left renal vein	Single vein passes **behind** aorta (≈ 2–3%)	Injured when dividing lumbar vessels / rolling aorta in retroperitoneal lymph node dissection (RPLND), aortic surgery
† Circumaortic left renal vein (renal collar)	Anterior + posterior limbs encircle aorta (≈ 5–7%)	Posterior limb runs caudally, may be mistaken for a lumbar vein; check pre-operative computed tomography (CT)
† Retrocaval (circumcaval) ureter	Right ureter passes behind IVC — a **venous** anomaly (persistent posterior cardinal vein)	"Reverse-J"/fish-hook on pyelogram; proximal obstruction → dismembered pyeloplasty anterior to IVC

6 · LYMPHATICS

- Lymph flows **cephalad and right → left** to the left innominate vein.
- Pelvic nodes → lumbar nodes → lumbar trunks → **cisterna chyli** (behind right side of aorta, in front of **L1–L2**) → thoracic duct, ascending behind aorta.
- Lumbar nodes are the primary drainage of the lateral-branch organs: kidney, adrenal, ureter, **gonad**.

Group	Extent
Left para-aortic (+ preaortic, retroaortic)	Aortic midline → left ureter; retroaortic between aorta and spine
Interaortocaval	IVC midline → aortic midline
Right paracaval (precaval, retrocaval)	IVC midline → right ureter; retrocaval between IVC and psoas

	Left testis	Right testis
Primary landing zone	**Left para-aortic**	**Interaortocaval**
Secondary	Interaortocaval	Right paracaval
Crossover	None significant to right	Small but real to left para-aortic

> **KEY POINT** Why the hilum? The testis is embryologically retroperitoneal — it drains to where it began. Crossover is one-way (right → left), following global lymph flow; hence right-sided templates include interaortocaval and some para-aortic tissue. → *Ch 80, 81*

7 · NERVES

Autonomic

- Two-neuron chain: preganglionic → ganglion → postganglionic. **Adrenal medulla** = modified sympathetic ganglion (preganglionic fibres synapse directly).
- **Parasympathetic**: craniosacral (III, VII, IX, X, S2–S4); ganglia in organ wall. **Sympathetic**: thoracolumbar **T1–L2**; trunk ganglia or splanchnics to aortic plexuses.
- **Sympathetic trunks**: medial to psoas on the anterolateral spine; lumbar vessels cross them at right angles — so trunks lie behind the great vessels and are met when lumbars are divided.
- Celiac → renal plexus (aorticorenal ganglion) supplies kidney, ureter, some testicular fibres. Superior hypogastric plexus: caudal aorta to front of L5.
- **Ejaculation**: † postganglionic fibres from the **T12–L3** sympathetic ganglia run anterior to the aorta/across the IVC and converge in the hypogastric plexus below the IMA; they mediate **seminal emission** and **bladder neck closure**. Disruption by extensive retroperitoneal dissection → loss of emission (anejaculation) or retrograde ejaculation. Erection (parasympathetic, pelvic) is unaffected. → *Ch 81*

Lumbar (lumbosacral) plexus

Anterior rami of T12, lumbar and sacral nerves; upper branches pierce psoas, lower run medial to it.

Nerve	Roots	Course / motor	Sensory
Subcostal	T12	Below 12th rib; wall muscles	Skin over hip
Iliohypogastric	L1	Over quadratus; internal oblique, transversus	Gluteal, pubic
Ilioinguinal	L1	As above	Upper medial thigh, root of penis, anterior scrotum
Genitofemoral	L1–L2	**On anterior surface of psoas**; genital branch → cremaster	Anterior scrotum/labium; femoral branch upper anterior thigh
Lateral cutaneous of thigh	L2–L3	Purely sensory	Lateral thigh
Obturator	L2–L4	Behind/medial to psoas → obturator canal; adductors	Medial thigh
Femoral	L2–L4	Lateral to psoas; quadriceps, iliacus, pectineus	Anterior thigh, medial leg (saphenous)

Nerve	Setting	Result
Genitofemoral	Psoas hitch sutures, laparoscopic varicocelectomy, RPLND on psoas	Anterior scrotal/upper thigh numbness
Obturator	Lateral-wall transurethral resection of bladder tumour (TURBT) (cautery reflex); pelvic node dissection	Adductor jerk → bladder perforation
Femoral	Retractor blade on inguinal ligament/psoas; low lithotomy	Loss of knee extension
Hypogastric plexus	Extensive retroperitoneal dissection	Anejaculation/retrograde ejaculation

8 · IMAGING ANATOMY

- CT axial sequence (cranial → caudal): upper poles above pedicles → renal arteries and veins (left renal vein crossing anterior to aorta) → renal pelves with **duodenum against right hilum** → lower poles and upper ureters.
- † Before RPLND/nephrectomy read CT for accessory renal arteries, retro-/circumaortic left renal vein and nodal size by zone.

9 · MCQ ONE-LINERS

1. D1 duodenum and pancreatic tail are NOT retroperitoneal.
2. Renal hilum at L1; right kidney lower.
3. Perirenal space is open inferiorly — collections track into the pelvis.
4. Lateroconal fascia separates anterior and posterior pararenal spaces.
5. Costovertebral ligament tethers 12th rib to L1–L2 transverse processes.
6. Gonadal arteries arise anterolaterally from aorta below renal arteries.
7. Marginal artery of Drummond allows IMA ligation.
8. Abdominal ureter supplied medially, pelvic ureter laterally.
9. Left gonadal vein enters left renal vein at a right angle.
0. New right varicocele → image the retroperitoneum.
11. IMV can be mistaken for gonadal vein at reoperation.
12. Circumaortic left renal vein: posterior limb mimics a lumbar vein. †
13. Retrocaval ureter is a venous (IVC) developmental anomaly. †
14. Cisterna chyli: right of aorta, anterior to L1–L2.
15. Right testis primary landing zone: interaortocaval.
16. Left testis primary landing zone: left para-aortic.
17. Lymphatic crossover: right → left only.
18. Emission fibres from T12–L3 converge in hypogastric plexus below IMA. †
19. Genitofemoral nerve lies on anterior surface of psoas.
20. Genital branch of genitofemoral is motor to cremaster.
21. Obturator reflex during lateral-wall TURBT risks bladder perforation.

79 Neoplasms of the Testis

Reading time ≈ 48 min (slow pace)

1 · EPIDEMIOLOGY AND RISK FACTORS

- >95% of testicular neoplasms are **germ cell tumours (GCTs)**: seminoma vs non-seminomatous germ cell tumour (NSGCT). 95% testicular, 5% extragonadal. Commonest solid malignancy in men **20–40 y**; peak 25–35.
- Cisplatin turned metastatic GCT from 5–10% to **80–90%** cure → the aim is now **minimum toxicity without losing curability**. Deaths come from platinum resistance and failure to clear residual disease early.
- Bilateral **2%**, mostly metachronous; histology discordant in 30–50%. Highest incidence in White men, Scandinavia; lowest in Black men, Africa/Asia; incidence rising.
- **Five established risk factors**: White race, cryptorchidism, family history, personal history, germ cell neoplasia in situ (GCNIS). Infertility also raises risk.
  - Undescended testis (UDT): relative risk (RR) **4–6**, falls to 2–3 if orchidopexy before puberty; contralateral descended testis RR ≈1.7.
  - Family history: brother RR 8–12 > father 2–4.
  - Previous GCT: 12-fold contralateral risk, yet 15-year cumulative incidence only 2%.
- **GCNIS** (= intratubular germ cell neoplasia): arrested fetal gonocyte formed **before birth**; adjacent to invasive GCT in 80–90%; risk of invasion **50% at 5 y**, 70% at 7 y. Contralateral GCNIS 5–9%, ≈36% with atrophy/UDT/age <40.
- Microlithiasis: significant only in men with previous GCT (marks GCNIS); alone in general population needs no work-up.

> **KEY POINT** GCT is a disease of the arrested gonocyte: UDT, family history and atrophy mark an abnormal developmental field, so orchidopexy reduces but never abolishes the risk.

2 · PATHOLOGY AND CLASSIFICATION

- World Health Organization (WHO) 2016†: postpubertal GCTs are **GCNIS-derived** (seminoma 52–56%; NSGCT 44–48% — embryonal carcinoma [EC], yolk sac tumour, choriocarcinoma, teratoma, mostly mixed) or **non-GCNIS-derived** (spermatocytic tumour, prepubertal teratoma/yolk sac tumour).
- **Any NSGCT element = NSGCT**, however small.
- **Isochromosome 12p, i(12p)**: in all postpubertal GCTs except spermatocytic tumour → proves GCT origin in carcinoma of unknown primary and in somatic-type malignancy from teratoma.

Subtype	Markers	Immunohistochemistry / histology	Hallmark
Seminoma	human chorionic gonadotrophin (hCG) in ≈15% (syncytiotrophoblasts); **never alpha-fetoprotein (AFP)**	CD117+, CD30−, placental alkaline phosphatase (PLAP) strongly +; clear cells, lymphocytic septa, granulomas	Older (4th–5th decade); can transform to NSGCT; "anaplastic" variant not recognised
Spermatocytic tumour	None	OCT3/4−, PLAP−, PAS−; three cell types	6th decade; not GCNIS, no i(12p), not with UDT, never mixed; **benign** (except sarcomatous/anaplastic variants)

Subtype	Markers	Immunohistochemistry / histology	Hallmark
Embryonal carcinoma	AFP and/or hCG, or normal	AE1/AE3+, OCT3/4+, PLAP+, CD30+†, c-KIT−	Totipotent; metastasises with normal markers; predicts occult spread in stage I
Yolk sac tumour	AFP, not hCG	Schiller–Duval bodies (≈50%), hyaline globules	Pure form: children, mediastinum
Choriocarcinoma	Very high hCG; no AFP	Syncytio- + cytotrophoblasts, haemorrhage	Haematogenous (lung, liver, brain); bleeds; usually poor risk
Teratoma	Normal or mild AFP	≥2 germ layers; mature/immature irrelevant postpubertally	Benign histology, malignant genetics; **chemoresistant**

- **Teratoma**: present in ≈half of mixed GCTs. Unresected → growing teratoma syndrome, unresectability, malignant transformation (rhabdomyosarcoma, adenocarcinoma, primitive neuroectodermal tumour — mostly at metastatic sites years after chemotherapy) or late relapse. This is why surgery remains integral to cure.
- hCG shares its α-subunit with thyroid-stimulating hormone (TSH) and luteinising hormone (LH) → hyperthyroidism, gynaecomastia.
- Expert pathology review changes subtype in ≈1/3 and lymphovascular invasion (LVI) status in ≈1/5.

> **KEY POINT** Management follows the least differentiated, most chemoresistant element: EC drives occult metastasis, choriocarcinoma drives haemorrhage, teratoma drives surgery.

3 · PRESENTATION

- **Painless testicular mass** commonest; acute pain from intratumoural haemorrhage/infarction (commoner in NSGCT). Trauma history is incidental.
- Metastases at diagnosis: ≈2/3 NSGCT vs 15% seminoma; metastatic symptoms present in 10–20% — back/flank pain (bulky nodes, ureteric obstruction), leg swelling (inferior vena cava), dyspnoea/haemoptysis, left supraclavicular node. Gynaecomastia ≈2%.
- Examine both testes, abdomen, inguinal nodes (after scrotal surgery), breasts, supraclavicular fossae.
- Differential: epididymo-orchitis (re-examine 2–4 weeks after antibiotics; persistent → ultrasound), torsion, haematoma, paratesticular tumour.
- **Delay**: ≈1/3 first misdiagnosed as epididymitis/hydrocele. Any man 15–50 with a firm testicular mass, midline retroperitoneal or left supraclavicular mass → testicular examination, ultrasound and markers (avoid wrong laparotomy/neck dissection).

4 · DIAGNOSIS AND INITIAL MANAGEMENT

Scrotal ultrasound (US)

- Extension of examination; 5–10 MHz. GCT is **hypoechoic**: homogeneous → seminoma; heterogeneous/cystic/calcified → NSGCT. Doppler flow supports malignancy; absence does not exclude. Scan both testes.
- Metastatic GCT with normal palpation → US for a **burned-out primary** (scar, coarse calcification) → orchidectomy (GCNIS/teratoma often persist). Normal testes → primary extragonadal GCT.
- Small (<10 mm) impalpable lesion, normal markers: 20–50% are GCT (mostly seminoma) → orchidectomy, inguinal exploration with frozen section, or serial US.
- Magnetic resonance imaging (MRI) if US equivocal.

Serum tumour markers

	AFP	hCG	Lactate dehydrogenase (LDH)
Source	EC, yolk sac	Choriocarcinoma, EC, syncytiotrophoblasts in seminoma	Non-specific; reflects bulk
Never from	Seminoma, choriocarcinoma	Yolk sac	—
Half-life	**5–7 days**	**24–36 h**	24 h
False positive	Liver disease, hepatocellular/gastric cancer, ataxia telangiectasia	LH cross-reactivity in hypogonadism (normalises 48–72 h after testosterone), marijuana, other cancers	Muscle injury, lymphoma

- Draw **before orchidectomy**, again after, during chemotherapy and in follow-up. Pre-orchidectomy values do not drive decisions; **post-orchidectomy values** (within 4 weeks of treatment) set S stage and risk group.
- Markers never decide whether to remove the testis. Raised AFP with "pure seminoma" histology → **treat as NSGCT**.
- Post-orchidectomy: decline by half-life → expected; slow decline with negative imaging → monitor; **rising → metastatic disease → chemotherapy**.
- Unstable patient with very high AFP/hCG: chemotherapy without tissue, orchidectomy afterwards (testis is a sanctuary behind the blood–testis barrier).
- miR-371a-3p: GCT-specific, not raised by teratoma; role still being defined.

Radical inguinal orchidectomy

- Testis + cord to internal ring via inguinal incision, within 1–2 weeks. **Trans-scrotal approach/biopsy contraindicated** — alters lymphatic drainage → inguinal/pelvic nodes, local recurrence.
- Curative alone in 80–85% of clinical stage (CS) I seminoma, 70–80% of CS I NSGCT.
- Report: each component and %, size, rete testis/tunica/cord invasion, pT, GCNIS, LVI, margin.
- **Testis-sparing surgery**: only for synchronous bilateral tumours or solitary testis (tumour <2–3 cm, adequate androgen) or suspected benign lesions; inguinal exploration → excision → frozen section → biopsies for GCNIS; GCNIS → ≥20 Gy to residual testis (may defer for paternity).
- Contralateral biopsy: consider if atrophic testis, UDT, age <40 or suspicious US.
- Extragonadal: raised AFP/hCG + normal testes is diagnostic (no biopsy); normal markers → biopsy; i(12p) supports GCT.

5 · STAGING

Spread and imaging

- Orderly lymphatic spread to retroperitoneum (first site in 70–80%) → distant; choriocarcinoma haematogenous. Cisterna chyli → thoracic duct → **left supraclavicular** node.
- **Landing zones**: right testis → **interaortocaval** below renal vessels; left → **para-aortic** and left renal hilar. Right → left crossover common; left → right implies bulk.
- Computed tomography (CT) abdomen–pelvis for all (also shows retroaortic renal vein, lower-pole artery before retroperitoneal lymph node dissection [RPLND]). False negative in **25–35% of CS I NSGCT** → landing-zone nodes 5–9 mm are suspicious; equivocal nodes with normal markers → repeat CT in 6–8 weeks.
- Chest X-ray for all; chest CT if markers raised, abdominal disease, abnormal X-ray, or CS I NSGCT with LVI/EC predominance. Isolated mediastinal nodes → biopsy (lymphoma/sarcoidosis).
- Brain imaging if hCG >10,000 IU/L. No routine bone scan. Fluorodeoxyglucose positron emission tomography (FDG-PET) no role at diagnosis.

- Offer **sperm cryopreservation** before RPLND, chemotherapy or radiotherapy (oligospermia in up to half at diagnosis).
- Base decisions on CT and markers done within 4 weeks of treatment, because GCT grows fast.

Tumour, node, metastasis (TNM) staging — the only TNM with a serum (S) category

Category	Definition
pTis	GCNIS
pT1	Testis/epididymis, **no LVI**; may invade tunica albuginea
pT2	With LVI, or into tunica vaginalis
pT3 / pT4	Spermatic cord / scrotum
N1 / N2 / N3	≤2 cm / >2–5 cm / >5 cm (pN also counts nodes: pN1 ≤5, pN2 >5 or extranodal; pN invalid after chemotherapy)
M1a / M1b	Non-regional nodes or lung / other viscera
S1	LDH <1.5×N *and* hCG <5000 mIU/mL *and* AFP <1000 ng/mL
S2	LDH 1.5–10×N *or* hCG 5000–50,000 *or* AFP 1000–10,000
S3	LDH >10×N *or* hCG >50,000 *or* AFP >10,000

Stage	Criteria
IA / IB	pT1 / pT2–4, N0 M0 S0
IS	Any pT, N0 M0, **S1–3** (raised markers, no radiological disease)
IIA / IIB / IIC	N1 / N2 / N3, M0, S0–1
IIIA	M1a, S0–1
IIIB	N1–3 or M1a with S2
IIIC	N1–3 or M1a with S3, or **any M1b**

International Germ Cell Cancer Collaborative Group (IGCCCG) risk — metastatic disease

Risk	NSGCT	Seminoma
Good	Testis/retroperitoneal primary, no non-pulmonary visceral metastases (NPVM), AFP <1000, hCG <5000, LDH <1.5×N (56%; 5-y survival 92%)	Any primary, no NPVM, any hCG/LDH, normal AFP (90%; 86%)
Intermediate	As above with any of AFP 1000–10,000, hCG 5000–50,000, LDH 1.5–10×N (28%; 80%)	Any primary **with NPVM** (10%; 72%)
Poor	**Mediastinal primary**, or NPVM, or AFP >10,000, hCG >50,000, LDH >10×N (16%; 48%, now ≈71%)	**None**

> **KEY POINT** IGCCCG uses markers at the start of chemotherapy, not pre-orchidectomy values, and is not valid at relapse. Seminoma risk depends on NPVM alone; mediastinal seminoma can be good risk, mediastinal NSGCT is always poor risk.

6 · TREATMENT PRINCIPLES

- Act fast, centralise care (outcome better at high-volume centres), give full-dose cisplatin despite cytopenias or renal impairment, resect all residual disease.

	Seminoma	NSGCT
CS I / II / III	85 / 10 / 5%	≈1/3 each
Occult metastasis in CS I	10–15%	25–35%
Radiosensitive	Exquisitely	No (except brain)
Metastatic teratoma	Rare	Frequent → surgery

	Seminoma	NSGCT
Markers	Rarely useful	Central

- Seminoma relapses contain NSGCT elements in 10–15% (30% at autopsy of seminoma deaths) → biopsy non-responders. Chemotherapy, radiotherapy and repeated CT all add late cardiovascular and SMN risk, whereas expert nerve-sparing RPLND has the most favourable long-term profile.

GCNIS management

- Options: orchidectomy, radiotherapy **≥20 Gy** (<20 Gy → recurrence), or observation. Radiotherapy kills germ cells but spares relatively radioresistant Leydig cells; still up to 40% later need testosterone.
- Subfertile man still suitable for assisted reproduction → serial US surveillance, or sperm retrieval and cryopreservation before orchidectomy.
- Normal contralateral testis wanting paternity → orchidectomy (radiotherapy scatter harms it). Cisplatin only partly clears GCNIS → biopsy ≥2 years later or radiotherapy.

7 · NSGCT MANAGEMENT

CS I NSGCT

- Survival ≈100% with any option; occult metastasis 20–30%, so any adjuvant treatment overtreats 70–80%.
- **Risk factors**: **LVI** (strongest) and **EC predominance**; high risk ≈50% relapse on surveillance, low risk <20%.
- **Surveillance**: relapse ≈28%, >90% within 2 y, 65–75% retroperitoneal; relapses are chemo-naive and curable. Disadvantages: CT radiation, long follow-up, non-compliance. Medical Research Council (MRC) trial: 2 vs 5 CTs equal.
- **Primary RPLND**: removes teratoma (15–25% of occult nodes); pathological stage II in 19–28%, most cured by RPLND alone; retroperitoneal recurrence <2% so no routine abdominal CT. Needs experienced surgeon; highest double-therapy rate. pN1 → observe; pN2–3 (relapse >50%) → **2 cycles adjuvant** (relapse ≤1%) or observe — same survival.
- **Primary chemotherapy, bleomycin–etoposide–cisplatin (BEP) ×1**: lowest relapse (≈2–3%), but leaves retroperitoneal teratoma, exposes everyone to late toxicity, and relapses are chemoresistant.
- Templates: full bilateral nerve-sparing (ejaculation >90%) is standard and mandatory for suspicious nodes; modified template acceptable for clinically negative nodes (right: may omit para-aortic below inferior mesenteric artery; left: may omit paracaval).

ALGORITHM
CS I NSGCT → low risk (no LVI, no EC predominance) → surveillance → high risk → surveillance or BEP×1 or nerve-sparing RPLND (expert) → choose by patient preference and toxicity

KEY POINT Decision analysis: surveillance maximises quality-adjusted survival when relapse risk <≈1/3. All three options cure ≈100%; they differ in how many are treated needlessly vs how hard relapse is to treat.

CS IS, IIA–B

- **CS IS** → induction chemotherapy as for advanced disease (RPLND leaves most needing chemotherapy anyway).
- **CS IIA–B**: raised markers or nodes **>3 cm** → induction chemotherapy (± post-chemotherapy RPLND). Normal markers, nodes <3 cm (especially teratoma in primary) → **primary RPLND**: 13–35% pN0 (CT false positive), ≈30% teratoma, half avoid chemotherapy. Indeterminate nodes → re-image in 6–8 weeks.

CS IIC–III: induction chemotherapy by IGCCCG

Risk / setting	Regimen	Note
Good	**BEP×3** or **EP×4** (etoposide–cisplatin)	BEP×3 = BEP×4; EP×3 inferior; carboplatin inferior
Intermediate / poor	**BEP×4**	VIP×4 (etoposide, ifosfamide, cisplatin) if poor lung function or major chest surgery planned
Second line	TIP×4 (paclitaxel, ifosfamide, cisplatin), VeIP×4 (vinblastine), VIP	TIP best for late relapse
Salvage	High-dose chemotherapy (HDCT, carboplatin–etoposide + stem-cell rescue)	Three negative randomised trials → specialised centres only

- Toxicities: cisplatin — nephro-, oto-, neurotoxicity; bleomycin — pulmonary fibrosis, Raynaud; ifosfamide — haematological, urological.
- Slow marker decline is adverse; intensification improved progression-free (PFS) but not overall survival → BEP×4 remains standard.
- HDCT rationale: dose escalation may overcome resistance; carboplatin replaces cisplatin because cisplatin's nephrotoxicity and neuropathy limit dose escalation. First-line HDCT failed in two randomised trials.

Post-chemotherapy residual masses (NSGCT)

- Normal markers + residual mass **>1 cm** → **post-chemotherapy surgery (PCS)/RPLND**, full bilateral template. Histology: necrosis ≈40%, teratoma ≈45%, viable ≈15%.
- Why always resect: no model, imaging or **FDG-PET** (teratoma not avid) predicts necrosis reliably (≈20% false negative); unresected teratoma → growth/transformation/late relapse.
- Complete responders (≤1 cm): observation or resection if pre-chemotherapy mass >3 cm — controversial.
- Multiple sites: resect all, retroperitoneum first; necrosis there predicts necrosis elsewhere.
- Viable malignancy: completeness of resection matters most; post-operative chemotherapy controversial. Somatic transformation (6–8%) → complete resection.
- Adverse factors after resection of viable tumour: IGCCCG intermediate/poor risk, incomplete resection, >10% viable cells. Inadequate PCS (mass-only or omitted) accounted for most relapses and deaths in the French BEP×3 vs EP×4 trial — surgery must be integrated properly.
- Marker-positive partial response or progression → second-line chemotherapy.

Relapsed NSGCT

- **Chemo-naive** (after surveillance/RPLND) → induction by IGCCCG, cure >95%; nonbulky marker-negative retroperitoneal relapse → RPLND an option.
- **Early post-chemotherapy (≤2 y)** → TIP/VeIP/VIP ×4 or HDCT; incomplete responders fare worst. Residual masses after salvage → post-salvage surgery (more viable tumour).
- **Desperation surgery**: rising markers but single resectable site → resect; long-term survival 33–57%.
- **Growing teratoma syndrome**: enlarging cystic masses with normal/falling markers during chemotherapy → stop chemotherapy and resect.
- **Late relapse (>2 y)**: ≈3%; yolk sac tumour commonest, often retroperitoneal, chemoresistant → **complete surgical resection**; TIP if chemotherapy needed. Biopsy if markers normal.

8 · SEMINOMA MANAGEMENT

CS I seminoma (≈80%)

	Surveillance (preferred)	Radiotherapy	Carboplatin
Detail	CT every 4–6 months for 3 y, then 6-monthly to 7 y, then yearly	Para-aortic (PA) strip 20 Gy (dog-leg [DL] if prior scrotal/inguinal surgery)	1–2 cycles, dose = 7 × (GFR + 25)

	Surveillance (preferred)	Radiotherapy	Carboplatin
Relapse	14–20%; retroperitoneal 84–100%	3–5%; thorax, supraclavicular; pelvis after PA	≈5% (×1), ≈1.5% (×2)
Drawback	Late relapses (10–20% after 4 y); CT burden	Second malignant neoplasm (SMN), gastrointestinal toxicity, oligospermia	Underdosing; unknown late effects

- Risk factors: tumour **>4 cm** and **rete testis invasion** (relapse 12/16/32% with 0/1/2) — **LVI is not predictive** in seminoma; not validated, so surveillance remains preferred for all.
- Calvert formula needs **isotope GFR** (glomerular filtration rate), not estimated GFR; otherwise give two cycles.
- Surveillance relapse: DL radiotherapy or chemotherapy (bulky/systemic); relapse after carboplatin treated as chemo-naive.
- Surveillance is harder than in NSGCT: CT detects >85% of relapses, hCG <5%, examination and chest X-ray almost none; median relapse 14 months (NSGCT 4–8) → long-term CT is mandatory.
- Radiotherapy late effects: peptic ulcer/dyspepsia, contralateral scatter oligospermia (8%), SMN ≈18% at 25 y; PA field chosen because pelvic nodes are involved in only 1–3% without prior scrotal/inguinal surgery.

CS IIA–B seminoma

- Nonbulky (≤3 cm): DL radiotherapy 25–30 Gy + 5–10 Gy boost, BEP×3/EP×4, or primary RPLND (SEMS, COTRIMS trials; ≈10% relapse, spares chemotherapy).
- Bulky >3 cm or multiple nodes → chemotherapy (radiotherapy relapse higher).

CS IIC–III seminoma

- Good risk (90%) → BEP×3 or EP×4; intermediate (NPVM) → BEP×4. Single-agent carboplatin inferior.

Post-chemotherapy residual mass (seminoma)

- Usually necrosis (≈90%); desmoplasia makes resection hard (complete in 58–74%, nephrectomy often). Many regress over 12–18 months.
- Viable tumour: >3 cm 13–55%, <3 cm 0–4%.

ALGORITHM
Residual seminoma mass → <3 cm → observe → >3 cm → FDG-PET ≥6 weeks after chemotherapy (earlier = inflammatory false positives) → PET-negative → observe → PET-positive → template post-chemotherapy RPLND (or repeat PET/close observation); mass-only excision → 33% local relapse

Relapsed seminoma

- After radiotherapy → first-line chemotherapy. After chemotherapy → VeIP×4 or HDCT; normal markers → biopsy (transformed NSGCT/teratoma possible). Late seminoma relapse may be favourable.

9 · BRAIN METASTASES

- ≈1% at diagnosis; linked to **choriocarcinoma/very high hCG**; risk of fatal haemorrhage during chemotherapy.
- At diagnosis: BEP×4 → resect residual lesions; radiotherapy/radiosurgery for unresectable residuals. Brain relapse after complete response is worse.

10 · TREATMENT SEQUELAE AND FOLLOW-UP

- RPLND: ejaculatory dysfunction (>80% if full template without nerve sparing), small-bowel obstruction. Orchidectomy alone: hypogonadism 10–20%.
- Chemotherapy: toxic death 0–2.4% (good risk), 3–4.4% (intermediate/poor); azoospermia almost universal, ≈half recover by 2 y, 80% by 5 y.

- Late: Raynaud 20–45%, neuropathy 14–43%, ototoxicity, hypogonadism 15–40%, infertility; **cardiovascular disease** (best documented; metabolic syndrome, vascular injury) and **SMN** (60–100% excess after cisplatin or radiotherapy, 200% after both; leukaemia, gastrointestinal, bladder). Surveillance CT adds radiation.
- Follow-up: markers, examination and imaging most intensive in years 1–2 (NSGCT) or 1–3 (seminoma); seminoma surveillance beyond 5 years; after carboplatin ≥5 years; after full RPLND chest imaging + markers (abdominal CT not routine). Survivorship care†: blood pressure, lipids, glucose, testosterone.

> **KEY POINT** Because cure is expected, late cardiovascular disease and second cancers — rising with each added modality — decide between equally curative options.

11 · NON-GERM-CELL TUMOURS†

- †**Leydig cell tumour**: commonest sex cord–stromal tumour (≈1–3% of testicular tumours); children → precocious puberty; adults → gynaecomastia, reduced libido (oestrogen excess). Reinke crystals; inhibin, calretinin, Melan-A positive; GCT markers normal. ≈10% malignant (size >5 cm, necrosis, LVI, high mitoses) — malignant forms resist chemotherapy and radiotherapy → RPLND for nodes.
- †Small, marker-negative lesions with endocrine features → testis-sparing excision with frozen section is reasonable.
- †**Sertoli cell tumour**: mostly benign; large-cell calcifying variant with Carney complex and Peutz–Jeghers syndrome (often bilateral; Peutz–Jeghers → gynaecomastia). Malignant features as for Leydig → orchidectomy ± RPLND.
- †**Lymphoma**: commonest testicular tumour in men **>50–60 y** and commonest bilateral testicular tumour; usually diffuse large B-cell. Orchidectomy for diagnosis → R-CHOP (rituximab, cyclophosphamide, doxorubicin, vincristine, prednisolone) + central nervous system prophylaxis + contralateral testis radiotherapy.

12 · LANDMARK TRIALS

Trial	Design	Result	Take-home
IGCCCG 1997	Pooled 5202 NSGCT, 660 seminoma	Markers, NPVM, mediastinal primary predict outcome	Risk groups select regimen
PVB vs BEP	RCT, advanced GCT	Equal survival, less neurotoxicity	BEP standard
BEP×3 vs ×4	RCT, good risk	Equal survival	BEP×3 for good risk
VIP vs BEP×4	RCT, intermediate/poor	Equal survival, more toxicity	VIP only if lungs at risk
MRC TE19/European Organisation for Research and Treatment of Cancer (EORTC) 30982	Carboplatin ×1 vs radiotherapy, CS I seminoma	5-y relapse-free 94.7% vs 96%	Carboplatin equivalent
MRC 20 vs 30 Gy	RCT, CS I seminoma	Equal relapse, less toxicity	20 Gy PA
Albers (German)	BEP×1 vs unilateral RPLND, CS I NSGCT	2-y PFS 99% vs 92%	Favours BEP×1; RPLND only by experts
Williams	Adjuvant ×2 vs observation, pathological stage II	Relapse 6% vs 49%; same survival	Both acceptable
MRC surveillance	2 vs 5 CTs, CS I NSGCT	No difference	Fewer scans safe

13 · MCQ ONE-LINERS

1. Commonest solid malignancy in men 20–40: GCT.
2. GCNIS: 50% invasive GCT at 5 years; forms before birth.
3. Only postpubertal GCT without GCNIS or i(12p): spermatocytic tumour — benign, sixth decade.
4. Raised AFP excludes pure seminoma and pure choriocarcinoma.
5. AFP half-life 5–7 days; hCG 24–36 hours.
6. Schiller–Duval bodies: yolk sac tumour.
7. Seminoma: CD117+, CD30−; EC: CD30+†, OCT3/4+.
8. Choriocarcinoma: very high hCG, haematogenous spread, brain haemorrhage.
9. hCG false positive in hypogonadism: LH cross-reactivity, corrected by testosterone.
0. Trans-scrotal orchidectomy alters drainage to inguinal nodes.
1. Right testis landing zone: interaortocaval; left: para-aortic/left renal hilar.
2. LVI upstages pT1 to pT2.
3. Stage IS: raised post-orchidectomy markers, no radiological disease → chemotherapy.
4. Any M1b = stage IIIC.
5. No poor-risk category for seminoma; mediastinal NSGCT always poor risk.
6. Brain imaging when hCG >10,000 IU/L.
17. CS I NSGCT occult metastasis predictors: LVI, EC predominance.
18. CS I seminoma relapse predictors: size >4 cm, rete testis invasion — not LVI.
19. Good-risk IGCCCG: BEP×3 or EP×4; intermediate/poor: BEP×4.
20. VIP replaces BEP when bleomycin lung toxicity must be avoided.
21. NSGCT residual mass >1 cm → RPLND; PET has no role.
22. Seminoma residual mass >3 cm → PET at ≥6 weeks.
23. Commonest histology in late relapse: yolk sac tumour.
24. Commonest somatic transformation in PCS specimens: rhabdomyosarcoma.
25. Growing teratoma syndrome: enlarging masses with normal markers → resect.
26. Carboplatin dose = 7 × (GFR + 25) using isotope GFR.
27. CS I seminoma standard: 20 Gy para-aortic if radiotherapy chosen.
28. GCNIS radiotherapy ≥20 Gy spares Leydig cells.
29. Commonest testicular tumour over 60 years: lymphoma†.
30. Reinke crystals: Leydig cell tumour†.
31. Large-cell calcifying Sertoli tumour: Carney complex, Peutz–Jeghers†.

80 Surgery of Testicular Tumours

Reading time ≈ 34 min (slow pace)

1 · EVALUATION OF THE TESTIS MASS

- Urgent: cancer is often rapidly progressive; delay escalates the intensity and morbidity of curative therapy.
- History (growth, symptoms) → examination (supraclavicular nodes, breasts, abdomen, then scrotum) → scrotal ultrasound (not mandatory; key role is the **contralateral testis** — bilateral synchronous ≈ **1%**) → serum tumour markers (STMs).
- STMs — α-fetoprotein (AFP), β-human chorionic gonadotropin (β-hCG), lactate dehydrogenase (LDH): support diagnosis of germ cell tumour (GCT), stage, and set the post-orchidectomy baseline.
- Discuss testicular prosthesis before surgery; † offer sperm banking before orchidectomy (subfertility common at diagnosis; contralateral testis may later be lost).

2 · RADICAL INGUINAL ORCHIDECTOMY

Diagnostic and therapeutic treatment of choice for suspected malignancy. Inguinal approach gives **early control of the cord** before the testis is handled, and removes testis, epididymis and cord to the internal ring. † A scrotal approach would breach a second lymphatic field (scrotal skin drains to inguinal nodes) and risk local recurrence.

ALGORITHM
Supine; prep umbilicus to mid-thigh and genitalia; EUA locates external ring → 3–5 cm transverse incision along Langer lines over the canal (hockey-stick extension if needed) → open external oblique ≈ 4 cm → **preserve ilioinguinal nerve** → mobilise cord at pubic tubercle, **encircle with Penrose (early clamp)** → divide external spermatic fascia/cremaster → deliver testis by upward scrotal pressure → divide gubernaculum → dissect cord to peritoneal reflection at internal ring.

- **High ligation**: vas and gonadal vessels ligated **separately** with **non-absorbable** suture, leaving a **1–2 cm** tail — marks the gonadal stump so it can be found and excised at later retroperitoneal lymph node dissection (RPLND). Vas, being separate, need not be removed at RPLND.
- Closure: irrigate → replace ilioinguinal nerve → close external oblique → subcutaneous → skin → scrotal support/fluffs 2–3 days (oedema, haematoma).
- † Prosthesis can be inserted at the same sitting through the inguinal wound and anchored in the dependent scrotum.

> **KEY POINT** † **Scrotal violation** (prior trans-scrotal orchidectomy/biopsy): complete with inguinal exploration, high cord ligation and excision of the scrotal scar; routine hemiscrotectomy is no longer needed (local recurrence low). Chemotherapy-era outcomes are not compromised.

Partial orchidectomy (testis-sparing surgery)

- **Indications (all three)**: polar tumour, **≤2 cm**, contralateral testis compromised/absent (synchronous bilateral, solitary testis). Justified only when anorchia risk outweighs local recurrence risk. **Never elective with a normal contralateral testis.**
- Uncertain malignancy → inguinal exploration, early cord control, excisional biopsy ± frozen section. Very small masses (**<5 mm**), negative markers → low malignancy rate; partial orchidectomy or, selectively, surveillance.
- Technique: as radical until delivery → isolate with towels → **intraoperative ultrasound** → cold ischaemia omissible if resection **<30 min** → incise tunica (vertical for ventral midline, horizontal for medial/lateral lesions — spares segmental arteries beneath tunica) → excise with rim of normal tubules → close tunica (absorbable) → three-point fixation.
- Surrounding germ cell neoplasia in situ (GCNIS): completion orchidectomy vs adjuvant radiotherapy **18–20 Gy** (abolishes spermatogenesis, preserves Leydig/androgen function) vs

observation (local recurrence ≈ 7.5%).

- **Ipsilateral local recurrence → completion radical orchidectomy**, always.

Delayed orchidectomy

Life-threatening metastatic GCT → chemotherapy first (diagnosis from biopsy or markers), then **delayed orchidectomy in all**: the testis is a sanctuary with **discordant chemotherapy response**. Specimens usually show teratoma or necrosis ("burned-out" primary). Apparent extragonadal retroperitoneal primary: biopsy shows intratubular neoplasia in up to 42%; nodes lateralising to a testicular pattern → orchidectomy on that side.

Post-orchidectomy staging: contrast-enhanced computed tomography (CT) abdomen/pelvis (magnetic resonance imaging [MRI] alternative); FDG-PET has no role in initial staging; repeat AFP, β-hCG, LDH — trend guides management.

3 · RPLND — TERMINOLOGY AND INDICATIONS

Term	Setting	Indication
Primary RPLND	No chemotherapy	High-risk clinical stage I or low-volume stage II (N1) nonseminomatous GCT (NSGCT), **normal STMs**
Post-chemotherapy (PC-RPLND)	After induction chemotherapy	Residual mass **>1 cm**, normal markers
Salvage PC-RPLND	After second-line chemotherapy	Residual disease
Desperation	Markers still elevated	Resectable residual disease
Redo	Prior RPLND	Retroperitoneal recurrence
Late relapse	**>24 months** after complete response	Retroperitoneal relapse

Each step down = more chemoresistant disease, harder surgery.

Preoperative planning

- CT within **6 weeks**, STMs within **7–10 days**; review vascular variants (duplicated renal vessels); no bowel prep; sperm banking offered.
- **Bleomycin lung**: pulmonary function tests → pulmonology → alert anaesthesia → **low FiO_2, restrictive fluids** → confirm at time-out.
- Inferior vena cava (IVC) thrombus: total occlusion → IVC resection; partial → reconstruction.

4 · TEMPLATES

Spread is predictable and side-specific: **right tumours → interaortocaval/precaval**; **left tumours → left para-aortic**. Contralateral and suprahilar spread rare, rises with bulk; interiliac very rare.

Template	Boundaries/content
Bilateral infrahilar	Renal vessels above, ureters laterally, common iliac arteries below; paracaval, precaval, interaortocaval, pre- and para-aortic
Modified right	Paracaval, precaval, interaortocaval, right common iliac; spares contralateral side †
Modified left	Para-aortic, pre-aortic, left common iliac; spares contralateral side †

- Consensus: omit **suprahilar** (↓ chylous ascites, renal and pancreatic/duodenal injury) and **interiliac** (+ contralateral) regions without bulky disease — preserves ejaculation.
- Against unilateral: 3–23% of pN+ have nodes outside modified template (right > left); some centres use full bilateral nerve-sparing for stage I/IIA. No randomised trial.
- **PC-RPLND standard = residual mass + full bilateral infrahilar** (crossover with bulk; viable tissue unrecognisable intraoperatively). Modified PC template only if **all three**: lesion **≤5 cm** in

landing zone, normal markers, International Germ Cell Cancer Collaborative Group (IGCCCG) good/intermediate risk.

5 · OPEN RPLND TECHNIQUE

- Supine, arms T; orogastric tube; xiphoid to below umbilicus. **Transperitoneal** (divide round, release falciform ligament) or **extraperitoneal** (peritoneum peeled off; bowel contained → faster bowel recovery, less ileus).
- Exposure: incise caecum tip to medial side of inferior mesenteric vein (IMV) → line of Toldt up to foramen of Winslow (bowel on chest) → free duodenum/pancreas off renal vessels; **liberal clips** to cut chyle leak.
- **Split and roll**: split at 12 o'clock on aorta below left renal vein → caudally to inferior mesenteric artery (IMA); ligate cephalad lymphatics at renal veins; roll packets off vessels; ligate lumbars.
- Lumbar arteries: usually 3 pairs; **2nd pair at IMA level** (landmark). Lumbar veins variable; the **common lumbar vein** is largest. At IMA level lie IMA, common lumbar vein and 2nd lumbar splanchnic nerve.
- Paracaval packet: smallest; spare genitofemoral nerve; **right sympathetic trunk** posterolateral to IVC is often mistaken for lymphatic tissue.
- Gonadal vein excised to internal ring and orchidectomy suture.

Nerve sparing

Emission and bladder-neck closure depend on **L1–L4 postganglionic sympathetic fibres** converging on the **superior hypogastric plexus** at/below the IMA, then over the sacral promontory.

	Left fibres	Right fibres
Location	Lateral aortic border, beneath left renal vein, around left common iliac artery	Emerge **beneath IVC**, run obliquely over aorta
Danger	Near gonadal artery and IMA	Beside superior and common lumbar veins — at risk when these are ligated

> **KEY POINT** Identify fibres **prospectively, before dividing anything**; sweep tissue off, loop them, then pass the packet beneath the nerve web. Avoid traction (neuropraxia caused transient anejaculation in early series).

Closure and postoperative care

- Clip lymph leaks; warm irrigation; optionally close posterior peritoneum and mesenteric root (volvulus).
- No routine drain (if large dissection: remove when serous, **<100 mL/24 h** on diet); no nasogastric tube; clear fluids day 0, low-fat diet days 1–2; non-opioid analgesia; mobilise day 1; discharge day 3–5.

6 · AUXILIARY PROCEDURES (PC-RPLND 23–45%)

- **Nephrectomy** (commonest): **left** (left hilar drainage); strongest predictor residual mass **≥10 cm**; risk rises fibrosis < teratoma < viable and in salvage/desperation/redo. Concern = future cisplatin eligibility.
- **Vascular**: plan in every PC-RPLND (subadventitial aortic injury → fatal haemorrhage). Small wall injury → pericardial patch; resection → polytetrafluoroethylene (PTFE)/Dacron graft.

	IVC	Aorta
Repair	Narrowing **<25%** → primary/lateral venorrhaphy; >25% → patch or graft	Small → primary; significant → graft
Ligation?	Yes (unstable bleeding; collaterals, sequelae rare)	**Never**

	IVC	Aorta
Note	Low pressure → grafts thrombose; mass >5 cm → ≈20% need IVC work	Duodenal serosal injury + graft repair → aortoduodenal fistula risk

- **Liver**: histology discordant with retroperitoneum in about half; marker-negative residual lesions → resect, or open microwave ablation if **<3 cm** unresectable.
- **Thorax**: 25–50% discordant → RPLND first; resect residual >1 cm unless retroperitoneum shows fibrosis only. **Pelvis**: mostly teratoma → resect completely.
- **Retrocrural**: transabdominal transdiaphragmatic incision along right crus, same sitting; extensive thoracic surgery → staged.

7 · DECISION MAKING

Complete remission after chemotherapy (≈70%)

Observation (Indiana and most centres) vs PC-RPLND for all (≤1 cm residual harbours teratoma ≈ 1/5, viable ≈ 1/20; untreated teratoma risks growing teratoma syndrome, late relapse, malignant transformation). Generally observe.

Predictors of necrosis only: no teratoma in orchidectomy specimen, normal pre-chemotherapy AFP/β-hCG, elevated LDH, >50% shrinkage, residual <2 cm.

Adjuvant chemotherapy after primary RPLND

Stage	Approach
pN0 / teratoma only	Observation
pN1	Observation (reliable patients)
pN2	Individualise: 2 cycles vs observation

Observation cures ≈70% by surgery alone; relapses salvaged but need ≥1 extra cycle of bleomycin, etoposide, cisplatin (BEP) or 2 extra of etoposide–cisplatin (EP).

8 · HISTOLOGY OF RESIDUAL MASS AFTER CHEMOTHERAPY

Finding	Frequency	Outcome/action
Fibrosis/necrosis	**≈40%**	relapse-free survival (RFS)/CSS ≈95%; no adjuvant
Teratoma	**≈45%**	RFS 80–90%; must be resected (chemoresistant, grows, transforms); no adjuvant
Viable GCT	**≈15%**	Survival 50–70%; consider 2 adjuvant cycles

RFS = recurrence-free survival; CSS = cancer-specific survival.

> KEY POINT **Why resect every NSGCT residual >1 cm?** No imaging (including PET †) or marker separates necrosis from teratoma; teratoma is chemoresistant and can grow (growing teratoma syndrome), relapse late or undergo malignant transformation. Surgery is therefore both the staging test and the only treatment for teratoma; histology decides further therapy.

Fizazi model (viable GCT): incomplete resection, ≥10% viable cells, IGCCCG intermediate/poor → 0/1/≥2 factors = 5-year OS 100/83/51%. Prognostic only.

Adjuvant: National Comprehensive Cancer Network (NCCN) — incomplete resection or intermediate/poor risk; European Association of Urology (EAU) — incomplete resection. Not after salvage chemotherapy.

9 · HIGH-RISK PC-RPLND

- **Salvage**: viable GCT at least doubled; OS 60–75%. After high-dose chemotherapy (HDCT): viable ≈40%, still worthwhile.

- **Desperation** (elevated markers): multidisciplinary only; OS ≈54%. Operate if disease confined (ideally retroperitoneum), completely resectable in one sitting, markers declining/plateauing, or chemotherapy exhausted. **Do not** with rapidly rising markers or multisite progression. Plateaued markers → >75% fibrosis/teratoma. **Post-op marker normalisation** is the only predictor of better OS.
- **Redo**: mostly technical failure — residual retroaortic/retrocaval tissue, incompletely divided lumbars. **Prevent by complete great-vessel mobilisation at first RPLND**; high-volume surgeons.
- **Late relapse**: yolk sac, somatic-type malignancy, chemoresistant → **surgery first** if resectable.

10 · UNIQUE SITUATIONS

- **Seminoma residual mass**: dense desmoplasia, no teratoma, viable ≈10%, more complications. **<3 cm** → observe; **>3 cm** → FDG-PET (≥6 weeks after chemotherapy †); PET-avid → PC-RPLND (resectable, select), salvage chemotherapy or HDCT.
- **Sex cord stromal tumours**: malignant features — size >4 cm, necrosis, mitoses, atypia, infiltration, lymphovascular invasion. Metastatic disease does badly; RPLND of unproven survival benefit.

11 · OUTCOMES, FERTILITY AND COMPLICATIONS

- Fertility: RPLND spares spermatogenesis; 40–60% have abnormal semen at diagnosis.
- **Ejaculation**: template alone or prospective nerve sparing → antegrade ejaculation **>90%** in primary RPLND; no oncological cost. † Anejaculation/retrograde ejaculation → sympathomimetics (pseudoephedrine, imipramine), post-ejaculatory urine sperm retrieval, electroejaculation or testicular sperm extraction.
- Overall complications: primary 10.6–24%; PC up to 56%. Mortality: primary ≈0, PC <1% (pulmonary embolism, duodenal fistula, MI, chylous ascites).

Complication	Key facts	Prevention/management
Pulmonary (bleomycin)	PC 3–5%; acute respiratory distress syndrome	Low FiO_2, restrictive fluids; tachycardia alone is not a fluid trigger; accept SpO_2 **88–90%**
Ileus	Up to 26% (commonest)	Extraperitoneal approach; no routine NG tube; early enteral intake
Chylous ascites	Primary 0.2–2.1%, PC 2–12%; suprahilar dissection disrupts cisterna chyli (L1–L2, behind aorta under left crus)	See algorithm
Lymphocele	Symptomatic ≤1.7%; air/rim enhancement = infected	Percutaneous drainage ± antibiotics
Vascular injury	Aorta, IVC, renal vessels	Vascular team available; repair per table above
venous thromboembolism (VTE)	Primary <1%, PC 0.1–3.1%	Compression devices, mobilise day 0, pharmacological if prior chemotherapy/VTE/obesity
Neurological	Femoral neuropraxia; paraplegia with bulky disease	Careful positioning/retraction

Chylous ascites: girth, anorexia, dyspnoea; tap → milky, protein >3 g/dL, triglycerides 2–8 × serum.

ALGORITHM

Paracentesis → capped drain if reaccumulates → very-low-fat, high-protein diet + medium-chain triglycerides + octreotide → nil by mouth + parenteral nutrition → refractory: lymphangiography ± embolisation, peritoneovenous shunt, or exploration → continue low-fat diet ≥**4 weeks** after resolution.

12 · LANDMARK TRIALS

Trial	Design	Result	Take-home
Williams (Testicular Intergroup)	randomised controlled trial (RCT), pathological stage II after RPLND: 2 adjuvant cycles vs observation	Adjuvant near-abolished relapse; observation relapses salvaged; equal survival	Observation acceptable for pN1; individualise pN2
SEMPET	Prospective FDG-PET in post-chemotherapy seminoma residuals	>3 cm: specificity 100%, NPV 96%; ≤3 cm positive predictive value (PPV) 37%	PET guides >3 cm masses; little role <3 cm

13 · MCQ ONE-LINERS

1. Suspected testis cancer → radical inguinal orchidectomy with early cord control.
2. Cord stump: non-absorbable suture, 1–2 cm tail, for retrieval at RPLND.
3. Bilateral synchronous testicular masses ≈ 1%.
4. Partial orchidectomy: polar, ≤2 cm, absent/compromised contralateral testis.
5. Cold ischaemia omissible if resection <30 minutes.
6. Adjuvant radiotherapy to GCNIS in remnant: 18–20 Gy.
7. Local recurrence after partial orchidectomy → completion radical orchidectomy.
8. Chemotherapy-first patients still need delayed orchidectomy (testis–metastasis discordance).
9. Primary RPLND requires normal post-orchidectomy markers.
0. Desperation RPLND = elevated markers after chemotherapy.
1. Late relapse: >24 months after complete response; treat surgically.
2. Right testis primary landing zone: interaortocaval.
3. Left testis primary landing zone: left para-aortic.
4. Second lumbar artery pair arises at the IMA level.
5. Right sympathetic fibres emerge beneath the IVC.
6. Postganglionic fibres L1–L4 converge on superior hypogastric plexus.
17. Commonest auxiliary procedure at PC-RPLND: left nephrectomy.
18. Strongest nephrectomy predictor: residual mass >10 cm.
19. IVC narrowing <25% → primary repair; aorta can never be ligated.
20. PC-RPLND histology: necrosis 40%, teratoma 45%, viable 15%.
21. Residual teratoma is chemoresistant; resect to prevent growing teratoma syndrome.
22. Chest residual: do RPLND first; retroperitoneal necrosis predicts thoracic necrosis ≈89%.
23. Commonest cause of redo RPLND: incomplete first surgery (retrocaval/retroaortic).
24. Seminoma residual <3 cm → observe; >3 cm → PET.
25. Seminoma residuals contain no teratoma.
26. Bleomycin patients: low FiO_2, restrictive fluids.
27. Commonest RPLND complication: ileus.
28. Chylous ascites triglyceride 2–8 times serum; risk ↑ with suprahilar dissection.
29. Low-fat diet ≥4 weeks after chylous ascites resolves.
30. Only predictor of OS after desperation RPLND: postoperative marker normalisation.
31. Leydig cell tumour: almost always stage I; Sertoli more aggressive.

81 Minimally Invasive Retroperitoneal Lymph Node Dissection

Reading time ≈ 17 min (slow pace)

1 · INDICATIONS AND RATIONALE

- Retroperitoneal lymph node dissection (RPLND) is central to testicular germ cell tumour (TGCT) treatment. Goals in early-stage disease: preserve cure, avoid **dual therapy** (surgery + chemotherapy), minimise lifelong toxicity.
- **Primary (chemotherapy-naive)**: marker-negative clinical stage I nonseminomatous germ cell tumour (NSGCT); small-volume stage II NSGCT and seminoma. Diagnostic + therapeutic — stages and cures occult disease, and spares chemotherapy in the **20–30%** with enlarged nodes who prove pN0.
- **Postchemotherapy (PC-RPLND)**: NSGCT residual mass **>1 cm**; select seminomas.
- Minimally invasive RPLND (MI-RPLND) is the latest de-escalation step, after dropping routine suprahilar dissection, nerve-sparing and modified templates.
- First laparoscopic RPLND MCQ 1992; robotic 2006. Robot adds wristed instruments, **fourth-arm retraction**, easier suturing and **circumferential great-vessel dissection with lumbar control**.

> **KEY POINT** Much recovery gain came from Enhanced Recovery After Surgery (ERAS) pathways, which help every approach — the benefit of access alone is smaller than it looks.

2 · GUIDELINES AND CONTROVERSY

Body	Position
National Comprehensive Cancer Network (NCCN)	Silent
American Urological Association (AUA), early-stage	**May be offered** by a surgeon with GCT and minimally invasive surgery (MIS) expertise; patient told long-term data are lacking
European Association of Urology (EAU)	Consider only in **specialist centres** expert in both open and MIS RPLND

Unlike nephrectomy, cystectomy and prostatectomy, there is no noninferiority evidence → the most controversial operation in urological oncology.

Concern	Basis / consequence
Staging-only dissection	No lumbar division, low node yields, excess adjuvant chemotherapy (even pIIA) → dual therapy; residual teratoma → late relapse
Wrong patients	Stage IA (should be observed) and many IB operated; pN0 up to 83%
Intraoperative safety	Great-vessel surgery from a console; robot-trained surgeons may lack open skills; conversion **≈5%**
Oncological uncertainty	No series > 5 years; peritoneal carcinomatosis reported

3 · PATIENT SELECTION

- No absolute contraindications; relative: extensive prior abdominal surgery, morbid obesity.

Suits MI-RPLND	Favour open
Chemotherapy-naive high-risk IB NSGCT electing RPLND	PC-RPLND in general; large residual mass
Small-volume II: 1–2 nodes in primary landing zone, none **>3 cm**	Large size change pre→post chemotherapy (desmoplasia)

Suits MI-RPLND	Favour open
PC ideal: solitary landing-zone mass, small response (suggests teratoma), little desmoplasia	Possible great-vessel reconstruction; somatic malignancy; retrocrural disease

4 · UNCHANGING OPERATIVE PRINCIPLES

- Multidisciplinary decision; imaging within **≈4 weeks**, markers within **1–2 weeks**.
- **MI-RPLND must completely duplicate open RPLND**: approach never changes completeness, template or nerve-sparing.

ALGORITHM
Safe entry → retroperitoneal exposure → **split-and-roll** complete nodal removal → prospective nerve identification/sparing → **lumbar vessel division** → spermatic cord removal → closure

Setting	Template
Stage I / small-volume II, chemo-naive	Extended unilateral (modified) with nerve-sparing — acceptable, debated
Enlarged nodes found at surgery	Convert to bilateral full template
Postchemotherapy	**Bilateral full template** with nerve-sparing = standard

- **Boundaries**: renal hila, ureters, ureter–common iliac crossing, plus spermatic cord. Para-aortic nodes above the inferior mesenteric artery (IMA) are in both modified templates.
- Lumbar arteries and veins are **intentionally divided** to clear retroaortic and retrocaval tissue; incomplete RPLND/mass excision worsens survival.
- **Nerve-sparing**: preserve sympathetic chains and L1–L4 postganglionic fibres (≈3 per chain) to their coalescence in the hypogastric plexus beneath the IMA, when oncologically safe.

5 · PREPARATION, POSITIONING AND PORTS

- Agreed haemorrhage plan; prior bleomycin → careful fluids and oxygen; crossmatch; **open set in theatre**; vascular sutures ready.
- Counsel: bleeding, anejaculation, visceral injury, chylous ascites, conversion. High-fat meal the day before highlights lymphatics; no full bowel prep.
- Orogastric tube; pad pressure points (rhabdomyolysis).

	Modified lateral decubitus (≈45°)	Supine, tilt (laparoscopic)	Steep Trendelenburg (robotic)
Ports	As renal surgery, slightly caudal	Four 12-mm midline trocars, first 1 cm below xiphoid	Camera 4–6 cm below umbilicus; two arms left, one right
Bowel retraction	Gravity + rotation	Gravity, tilting	Bowel falls cranially; **peritoneal hammock**
Note	Familiar; bilateral needs repositioning	Bilateral, no repositioning	Any template

Hammock: posterior peritoneum incised from Treitz around caecum; edges sutured to anterior wall to suspend bowel.

6 · OPERATIVE STEPS

Right template

1. Incise line of Toldt from caecum round hepatic flexure to foramen of Winslow; **Kocherise** duodenum.
2. Separate ureter from cord; divide gonadal vein at inferior vena cava (IVC) ostium.

3. **Split** over right common iliac at ureteric crossing, caudal → cranial along IVC to the left renal vein; clip retropancreatic/retroduodenal lymphatics.
4. **Roll** paracaval packet laterally → find sympathetic chain lateral to lumbars → spare fibres → divide lumbars (Weck clips on vessel side, sealer on specimen side) → free off psoas.
5. Interaortocaval packet rolled medially off the anterior spinous ligament. **Do not ligate the right renal artery** here (often at lower edge of left renal vein); preserve accessory lower-pole arteries.

Left template — differences

- Divide gonadal vein at left renal vein; control the **lumbar vein entering the left renal vein**; clip the lymphatic-rich tissue around it.
- Split along aorta from left common iliac to left renal vein; roll laterally; IMA may be ligated to aid mesocolic rotation.

Closure

Lower pneumoperitoneum to check haemostasis/lymphostasis → bag packets promptly (avoid spill) → **no routine drain**. Post-op: early diet and ambulation, non-narcotic analgesia; discharge **1–2 days**.

7 · COMPLICATIONS AND MANAGEMENT

Major vascular injury (most feared)

ALGORITHM
Pressure (instrument/assist port) + raise pneumoperitoneum → prepare: inform anaesthesia, blood, vascular help → if uncontrolled: keep pressure, undock, convert

- If repaired MIS: extra trocar, sponges; **identify the injury before acting — no blind clipping**.

Chylous ascites

- Prevent: liberal clips at left renal vein and inferior mesenteric vein crossing (energy devices seal lymphatics unreliably); low-pressure check; sealants. Rates match open surgery.
- † Treat: low-fat/medium-chain triglyceride diet → paracentesis, octreotide, total parenteral nutrition → surgical ligation if refractory.

Ejaculatory dysfunction

- Antegrade ejaculation preserved in **≥90%** after primary nerve-sparing MI-RPLND.

8 · OUTCOMES VS OPEN

- Case series only; **no randomised trial**.
- Shorter stay, less blood loss; longer operative time. Complications match open; brisk bleeding is the unique risk.
- Recent robotic node yields approach the open benchmark **≈30**; adjuvant chemotherapy falling but still too frequent.
- Benchmark: recurrence-free ≈**85%** for limited pN+ without adjuvant chemotherapy.
- Low MI recurrence is **confounded** by high pN0 rates and adjuvant chemotherapy — not proof of equivalence. PC-RPLND data are too small for conclusions.
- Future: **miR-371a-3p** to select who needs RPLND; RPLND for stage II seminoma.

9 · LANDMARK TRIALS

Trial	Design	Result	Take-home
GETUG T93BP	Good-risk NSGCT: bleomycin, etoposide, cisplatin (BEP) ×3 vs etoposide–cisplatin (EP) ×4	Most relapses/deaths in those without or with incomplete RPLND	Complete RPLND matters

10 · MCQ ONE-LINERS

1. PC-RPLND in NSGCT: residual mass >1 cm.
2. 20–30% with enlarged nodes are pN0 at RPLND.
3. First laparoscopic RPLND 1992; first robotic 2006.
4. NCCN is silent on MI-RPLND.
5. EAU: MI-RPLND only in specialist centres.
6. Ideal small-volume II for MIS: 1–2 landing-zone nodes, none >3 cm.
7. Large chemotherapy shrinkage predicts desmoplasia → favour open.
8. Post-chemotherapy standard: bilateral full template with nerve-sparing.
9. Lumbar vessels divided to clear retroaortic/retrocaval nodes.
0. Nerve-sparing preserves L1–L4 postganglionic fibres to hypogastric plexus.
11. Robotic MI-RPLND: steep Trendelenburg with peritoneal hammock.
12. Right template mobilisation includes Kocher manoeuvre.
13. High-fat meal pre-op highlights lymphatics.
14. Weck clips on great-vessel side of lumbar vessels.
15. Do not ligate right renal artery in interaortocaval space.
16. Left template: control lumbar vein into left renal vein.
17. Conversion rate ≈5%.
18. Antegrade ejaculation preserved ≥90% after primary MI-RPLND.
19. Open RPLND node-yield benchmark ≈30.
20. miR-371a-3p may select RPLND candidates.

82 Tumours of the Penis

Reading time ≈ 48 min (slow pace)

1 · OVERVIEW

- Squamous cell carcinoma (SCC) = **~95%** of penile malignancy. Rare in North America/Europe (**0.4–0.6%** of male cancers); up to **10%** in parts of Asia, Africa, South America.
- Disease of older men (mean mid-50s, abrupt rise in 6th decade), but a meaningful minority are <40 years.
- Prognosis and planning rest on **pathological stage and histology of the primary** and on **presence and extent of nodal metastasis**.

2 · PREMALIGNANT LESIONS — PENILE INTRAEPITHELIAL NEOPLASIA (PEIN)

- **PeIN** = dysplastic penile squamous epithelium with an **intact basement membrane** (formerly carcinoma in situ, CIS).
- 2022 World Health Organization (WHO): **all PeIN is high grade** — PeIN I/II/III, "low/high-grade dysplasia" and "squamous CIS" are abandoned as pathological terms.
- Divided, like invasive SCC, by human papillomavirus (HPV) status:
  - **HPV-associated PeIN** (commonest; basaloid and warty subtypes) — precursor of HPV-associated SCC.
  - **HPV-independent (differentiated) PeIN** — hyperplastic epithelium, keratin pearls, prominent intercellular bridges, atypical basal cells; hard to separate from reactive (pseudoepitheliomatous) hyperplasia; associated with lichen sclerosus (LS).
- **Mechanism**: high-risk HPV (hrHPV) E6 binds p53, E7 binds retinoblastoma protein (Rb) → cell-cycle progression, deranged apoptosis. Rb inactivation → **p16INK4a (p16) overexpression** → p16 immunohistochemistry is the routine surrogate of hrHPV.

Clinical entity (all = PeIN)	Site / appearance	Progression if untreated
Erythroplasia of Queyrat (EQ)	Glans (± prepuce), uncircumcised; red, velvety, well-marginated; may ulcerate. Mimic: chronic balanitis	**10–33%**
Bowen disease (BD)	Shaft; sharply defined scaly erythematous plaque. Mimics: psoriasis, eczema, superficial basal cell carcinoma	**~5%**
Bowenoid papulosis (BP)	Shaft, young men; papules; HPV-related	**Does not progress**

- Eponyms describe site and look only — no unique histology. EQ vs balanitis: atypia at all levels, bulbous rete, submucosal capillary proliferation, plasma-cell infiltrate. Metastasis from PeIN extremely rare.
- **Giant condyloma (Buschke–Löwenstein)**: HPV-related (6/11), locally destructive, non-metastatic, but harbours coexisting SCC → complete excision + follow-up. Imiquimod is the topical treatment of choice for condyloma.
- Penile Kaposi sarcoma (human herpesvirus 8) → test for human immunodeficiency virus (HIV)/immunosuppression.

Treatment of PeIN — eradication with organ preservation

- **Multiple biopsies of adequate depth first** to exclude invasion.
- Foreskin: circumcision / excision to negative margin.
- Glans: topical **5-fluorouracil (5-FU)** or **5% imiquimod**; laser (neodymium:yttrium-aluminium-garnet [Nd:YAG], potassium titanyl phosphate [KTP], CO_2); **glans resurfacing** (very low recurrence).

3 · AETIOLOGY AND RISK FACTORS OF SCC

Factor	Key point
No neonatal circumcision	Neonatal circumcision virtually eliminates invasive cancer (rare in Jewish men). Delayed (puberty/adult) circumcision gives little/no protection — exposure already occurred. Protects against invasive SCC but **not CIS**
Phimosis	Present in **25–75%**; **7–10×** risk. The non-circumcision risk is mediated by phimosis. Cycle: poor hygiene → smegma → inflammation → fibrosis → phimosis
HPV	Low risk 6, 11 (warts, verrucous). High risk **16, 18**, 31, 33, 45, 52, 58; **HPV 16 commonest** (also in metastases). Present in only **~50%** of SCC; tracks number of sexual partners; wives have ~3× cervical cancer risk
HIV	More HPV; ~4× risk of penile cancer death; progression from PeIN ~6 years earlier
Tobacco	All forms (smoking, chewing, snuff); dose-dependent, independent of phimosis
Lichen sclerosus (balanitis xerotica obliterans, BXO)	Cancer in **2.3–9%** on long follow-up; specifically HPV-**independent** SCC
Trauma, obesity	Scarred shaft after mutilating circumcision; obesity → buried penis, functional phimosis
† **Psoralen + ultraviolet A (PUVA)**	† Genital PUVA exposure (psoriasis) markedly increases penile SCC → shield genitalia

- HPV-independent pathway: CDKN2A promoter hypermethylation, TP53/p14ARF mutations, MDM2 overexpression; inflammation (LS, phimosis, balanitis) → reactive oxygen species, cyclo-oxygenase-2 (COX-2) → DNA damage, proliferation.
- **Not linked**: occupation, gonorrhoea, syphilis, herpes, alcohol, marijuana.
- **Prevention**: neonatal circumcision (American Academy of Pediatrics 2012: benefits outweigh risks), hygiene, avoid tobacco, HPV vaccination (bivalent 16/18; quadrivalent 6/11/16/18; nonavalent adds 31/33/45/52/58).

> **KEY POINT** Two routes converge on the preputial sac: HPV-driven (E6/E7, p16+, basaloid/warty) and inflammation-driven HPV-independent (phimosis, smegma, LS). Neonatal circumcision, hygiene and vaccination each prevent part; adult circumcision prevents almost none. Penile cancer is the only neoplasm with a simple prophylaxis that spares the organ.

4 · NATURAL HISTORY AND SPREAD

- Starts on glans (~½), prepuce, coronal sulcus; shaft <2%. Papillary/exophytic or flat/ulcerative — **flat ulcerative tumours metastasise earlier**, poorer survival.
- **Buck fascia** is a temporary barrier; penetration of it and tunica albuginea → corporal invasion → vascular spread.
- **Lymphatics (stepwise, bilateral)**: prepuce/shaft skin → superficial inguinal nodes (above fascia lata); glans/corpora → superficial → deep inguinal (below fascia lata) → pelvic (external iliac, internal iliac, obturator). Cross-connections at every level → drainage is bilateral.
- Distant metastases (lung, liver, bone, brain) uncommon (**1–10%**), late; distant without nodal disease unusual. Untreated: death within **2 years** (femoral erosion haemorrhage, sepsis, inanition); no spontaneous remission.

5 · PRESENTATION AND DIAGNOSIS

- Lesion: induration, papule, warty growth, exophytic mass or ulcer with rolled edges; may be hidden by phimosis (foul discharge, bleeding). Pain disproportionately slight.

- **Delay**: 15–50% wait >1 year (embarrassment, fear); physician delay from prolonged antibiotic/antifungal trials.
- **Examination**: size, site, fixation, corporal involvement, penile base, scrotum; rectal/bimanual (perineal body); **careful bilateral inguinal palpation**.
- **Biopsy mandatory before definitive therapy** — confirms carcinoma, depth and grade; predicts occult nodal risk in clinically node-negative (cN0). Dorsal slit for exposure, or frozen section then immediate resection. Superficial biopsies mislead (depth indeterminable, grade discordant).
- **Labs**: anaemia, leucocytosis, hypoalbuminaemia; **hypercalcaemia without bone metastases** (PTH-related substances; linked to bulky inguinal disease; may resolve after node excision). Manage: saline hydration → bisphosphonate (pamidronate, zoledronic acid) → calcitonin for rapid lowering; diuretic only if overloaded.
- **Differential**: condyloma, Buschke–Löwenstein, BXO, chancre, chancroid, herpes, lymphogranuloma venereum, granuloma inguinale, tuberculosis.

6 · HISTOLOGY

Subtype (WHO 2022)	HPV	Features	Nodal risk / behaviour
Usual (keratinising) **45–65%**	Independent	Usually grade II; lymphovascular (LVI) or perineural invasion (PNI) in ~⅓	25–40%
Warty (condylomatous)	Associated	Cauliflower, fibrovascular cores, jagged border; slow, bulky	17–25%
Basaloid	Associated	Small basaloid cells, central necrosis, deeply invasive	**>½ node positive**
Papillary	Independent	Low grade, no koilocytosis, jagged border	≤12%
Verrucous	Independent	Very well differentiated, **broad pushing border**	**No metastatic potential**; recurs if under-excised
Sarcomatoid	Independent	Spindle component; mimics sarcoma/melanoma	**Most aggressive**
Rare: pseudoglandular, adenosquamous (HPV-independent); clear cell, lymphoepithelioma-like (HPV-associated)	—	Deep corporal invasion	Pseudoglandular, clear cell worst

- **Grading**: 3-tier WHO/International Society of Urological Pathology (ISUP); **any proportion of anaplastic cells = grade III** (half of tumours are heterogeneous, and any grade III component predicts nodes). Broders grading (1921) historic.
- **Vascular invasion** predicted nodal metastasis in every study; **PNI** a strong predictor. Synoptic reporting: grade, structure invaded, LVI, PNI, margins.
- **Molecular**: **hrHPV/p16 positivity = better disease-specific survival** (predicts survival, *not* nodal status). Adverse: p53, programmed death-ligand 1 (PD-L1), raised C-reactive protein (CRP), neutrophil-to-lymphocyte ratio (NLR). Nomograms unvalidated.

7 · IMAGING

Target	Approach
Primary	Examination reliable, esp. small distal lesions. Ultrasound sees tunica albuginea (good for cavernosal invasion) but cannot separate T1/T2. **Magnetic resonance imaging (MRI)** ± artificial erection (intracorporeal prostaglandin E1) when examination equivocal and organ sparing planned. Computed tomography (CT) not useful
cN0 groin	**No imaging detects micrometastases** (CT, PET, ultrasound-guided fine-needle aspiration cytology [FNAC] all insensitive) → surgical staging. CT if obese or prior groin surgery
cN+ groin / T3–T4	CT chest/abdomen/pelvis; central necrosis or irregular nodal border → poor pathology (≥3 nodes, extranodal extension [ENE], pelvic nodes). **Fluorodeoxyglucose positron emission tomography (FDG-PET/CT)** best for pelvic and distant disease

8 · STAGING — TUMOUR-NODE-METASTASIS (TNM) 8TH EDITION (AMERICAN JOINT COMMITTEE ON CANCER/UNION FOR INTERNATIONAL CANCER CONTROL, 2017)

T	Definition
Tis	PeIN
Ta	Non-invasive localised SCC (pure verrucous without destructive invasion; non-invasive papillary/warty/basaloid)
T1	Glans: lamina propria · Foreskin: dermis/lamina propria/dartos · Shaft: connective tissue between epidermis and corpora
T1a	**No LVI, no PNI, not high grade**
T1b	**LVI and/or PNI, or high grade (grade III/sarcomatoid)**
T2	**Corpus spongiosum** (glans or ventral shaft) ± urethra
T3	**Corpora cavernosa** (incl. tunica albuginea) ± urethra
T4	Scrotum, prostate, pubic bone

N	Clinical (cN)	Pathological (pN)
1	Palpable mobile unilateral node	**≤2 unilateral** inguinal metastases, no ENE
2	Mobile ≥2 unilateral or bilateral	**≥3** unilateral or bilateral
3	Fixed inguinal mass or pelvic nodes	**ENE or pelvic nodes**

Stage	I	IIA	IIB	IIIA / IIIB	IV
TNM	T1a N0	T1b, T2 N0	T3 N0	T1–3 N1 / N2	T4, any N3, any M1

- Changes from 7th: PNI added to T1b; T2 (spongiosum) vs T3 (cavernosa) split because cavernosal invasion has worse survival; **urethral invasion no longer upstages**; pN1 widened to 2 nodes (1 vs 2 nodes: equal survival) — fewer men need adjuvant therapy.
- Work-up: T = examination + biopsy/resection; N = examination, CT if palpable, dynamic sentinel node biopsy (DSNB)/surgical staging if high risk; M = liver function, **calcium**, CT chest/abdomen/pelvis, bone scan or FDG-PET/CT as indicated.

> **KEY POINT** Traps: T1b and T2 share stage IIA; any N3 or T4 = stage IV even without metastases; cN counts palpable nodes, pN counts metastases (pN1 allows two). The primary is staged by tissue, the groin by surgery.

9 · MANAGEMENT OF THE PRIMARY TUMOUR

- **Penectomy** = oncological gold standard (local recurrence **0–8%**) but harms sexual function and psyche; >half are ≤60 years → penile preservation where safe.

- **Organ preservation suits**: PeIN, Ta, T1, grade I–II in **compliant** patients. **Against**: size **≥4 cm**, grade III, deep cavernosal invasion, unfit for salvage.
- Goals: preserve glans sensation if possible; otherwise maximise shaft length.
- Local recurrence is higher than after amputation, but **early-detected recurrence is salvaged without survival cost** — except in pT3/high-grade, where local relapse independently worsens survival.

Margins — the 2-cm dogma is abandoned

- Microscopic proximal spread tracks grade: **~5 mm** (grade I–II), **~10 mm** (grade III); no skip lesions.
- Only margins **<1 mm** raised recurrence → a **negative margin (>1 mm) on frozen section is enough; margins well under 1 cm are acceptable** for small low-grade tumours.

Stage / site	Treatment (Table 82.4)
PeIN, glans	Laser, glans resurfacing; topical alternative
Ta / PeIN — foreskin or shaft skin	Excision to negative margin; laser; topical (Tis only)
Ta, T1 (grade I–III) — glans	Excision, resurfacing, glansectomy, radiotherapy (RT; not for Ta)
T2 glans, no gross cavernosal involvement	**Total glansectomy** ± distal corporal transection; partial penectomy; RT
T2 with corporal invasion, T3	**Partial or total penectomy**
T4	Neoadjuvant chemotherapy → surgery in responders
Local recurrence after conservation	Re-excision; partial/total penectomy; superficial low-grade may be re-treated conservatively

Techniques

- **Topical 5-FU / imiquimod**: Tis only.
- **Glans resurfacing** (glans CIS): document no invasion → strip epithelium and subepithelial tissue off spongiosum (total better than partial — fewer positive margins) → split-skin graft; topical therapy for isolated CIS at margin. Erections preserved; some need re-excision for unexpected invasion.
- **Laser**: CO_2 penetrates ~0.1 mm (too superficial; recurrence up to 50%); **Nd:YAG** (1064 nm, up to 6 mm) preferred, with biopsies of the tumour base. Recurrence <10% (CIS), 10–25% (T1) → **PeIN is the ideal laser cohort**.
- **Mohs micrographic surgery**: layer-by-layer margin control; modern frozen-section Mohs recurrence ~⅓ → no clear advantage over excision with frozen section.
- **Glansectomy** (± distal corporal transection) with skin graft neoglans — select low-stage, low-grade.
- † **Partial penectomy**: tourniquet at base; transect corpora with negative margin; spatulated neomeatus; aim for a stump allowing standing micturition. † **Total penectomy** when the residual stump would be inadequate or tumour is proximal: divide corpora at crura, **perineal urethrostomy** (seated voiding).

> **KEY POINT** Margin width is a function of grade, not a fixed 2 cm. Low-grade tumours spread only millimetres and their recurrences are salvageable — so preservation is safe. In high-grade or cavernosal disease, local relapse itself becomes lethal, and amputation regains its place.

10 · INGUINAL NODES — BRIEF ALGORITHM *→ CH 84*

- **Presence and extent of inguinal metastasis is the most important prognostic factor.** Lymphadenectomy alone can cure (prolonged locoregional phase).
- Cure ~**80%** if ≤2 nodes, unilateral, no ENE, no pelvic nodes; pelvic nodes positive → ~**14%** 5-year survival with surgery alone.

- **Early vs delayed**: node-positive men found by early staging survive far better than those dissected after groin relapse on surveillance (ENE almost universal when delayed) → stage early.
- Palpable nodes: ~half are inflammatory, but **no 4–6-week antibiotic trial** now — **FNAC** at presentation. Negative FNAC does not exclude → repeat or excisional biopsy.
- Occult nodal risk by primary: PeIN/Ta/T1a **0–18%** (low risk); **T1b or above** one-third to >half (high risk). Corporal invasion ~60%.

ALGORITHM

cN0, low risk (PeIN, Ta, T1a) → surveillance (compliant) → palpable node → FNAC/core → positive → ipsilateral inguinal lymph node dissection (ILND) + contralateral staging.

cN0, high risk (T1b+, LVI, PNI, grade III) → bilateral **DSNB** (preferred; ultrasound ± FNAC first) or superficial / fascia-sparing / minimally invasive ILND → positive side → completion ILND.

cN1–cN2 mobile, <4 cm → biopsy → fascia-sparing or radical ILND + contralateral staging; pelvic lymph node dissection (PLND) if >2 nodes or ENE.

Fixed / ≥4 cm / pelvic nodes → cisplatin-eligible: neoadjuvant TIP → resect if response/stable; ineligible: chemoradiation → resect if response.

Adjuvant if >2 nodes, bilateral, ENE or pelvic nodes.

- **DSNB**: blue dye + radiocolloid gamma probe; refinements (pre-op ultrasound ± FNAC, serial sectioning + cytokeratin, exploration of non-visualising groins) cut false-negatives from ~18% to ~7%. Lowest morbidity. Anatomical (Cabanas) sentinel biopsy abandoned.
- **Morbidity** (skin necrosis, infection, seroma, lymphoedema) tracks the amount of tissue removed: radical > superficial/fascia-sparing > minimally invasive > DSNB. Reduce it with transverse incisions, saphenous vein sparing, drain removal by volume, thromboprophylaxis; sartorius transposition *increases* complications.
- No pelvic skip metastases if inguinal nodes negative. Positive pelvic nodes are best treated by neoadjuvant chemotherapy then surgery; PLND is mainly a staging tool selecting adjuvant therapy. Follow-up of observed groins: low risk 3-monthly, high risk 2-monthly for years 1–2 (most relapses).

11 · RADIOTHERAPY

Primary

- Penile-preserving with curative intent; surgical salvage remains curative. **Circumcise first** — exposes lesion, settles infection, prevents preputial oedema and later phimosis.
- **External-beam RT (EBRT)**: **≥66 Gy in 2-Gy fractions over ~6.5 weeks**; penis vertical in a wax/Perspex block that acts as **bolus** (megavoltage beams spare skin, but tumour is cutaneous). ~55% recurrence-free at 5 years. Better control: dose >60 Gy, overall time ≤45 days, fraction ≥2 Gy; worse: T3, >4 cm, high grade. For unfit or locally advanced patients (± weekly cisplatin).
- **Brachytherapy** (iridium-192 interstitial needles; low [LDR], pulsed or high dose rate): for **T1–T2 glans tumours <4 cm**; ~60 Gy over ~5 days (LDR); ~79% relapse-free, penile preservation ~85% at 5 years; better erectile preservation than EBRT; salvage more often partial than total penectomy. Toxicity: meatal stenosis (dose >60 Gy, volume >25 cc), ulceration (dose >65 Gy). Advantage partly selection.
- Surface mould: superficial/non-invasive tumours only (rapid depth fall-off).
- **Late effects**: **meatal stenosis** (10–45%, 18–24 months; self-dilatation) and **ulceration** (flat, no raised edge; conservative care, hyperbaric oxygen; biopsy if suspicious). Up to **1 in 5 local recurrences occur after 5 years** → long follow-up.
- Nodal staging indications are **identical** after RT or surgery to the primary (biopsy supplies grade/LVI).

Nodal regions

- **No prophylactic groin RT** for high-risk cN0 — surgical staging instead (unless unfit).

- Adjuvant groin (± pelvic) RT for **≥2 positive nodes or ENE** (extrapolated from vulval cancer): ≥50 Gy, **54 Gy for ENE, 57–60 Gy for gross residual**; ≤50 Gy → more in-field failure. HPV-positive patients may gain more. Chemoradiation (cisplatin if glomerular filtration rate >50 mL/min) can convert or treat inoperable nodes.

12 · SYSTEMIC THERAPY

- No completed randomised trials. Most active: **platinum + taxane triplets — paclitaxel, ifosfamide, cisplatin (TIP)**; docetaxel, cisplatin, 5-FU (TPF; toxic). Doublets (cisplatin + 5-FU, paclitaxel + carboplatin) for less fit.
- **Neoadjuvant TIP** (4 cycles) for **bulky (N2–N3) inguinal or any pelvic nodes**: response ~50%; long-term disease-free survival in ~⅔ of responders vs very few non-responders → **response selects for consolidative ILND/PLND**. Non-responders rarely cured by surgery.
- **Adjuvant chemotherapy**: no proven overall survival gain; consider in pN3/pelvic nodes.
- **Bleomycin regimens discouraged** (vincristine–bleomycin–methotrexate historical) — unacceptable toxicity.
- Palliative: platinum first line (~3× median survival vs supportive care — months); TIP/TPF unsuitable with neuropathy or low GFR. Second line (vinflunine, paclitaxel) median survival ≤6 months. Anti-epidermal growth factor receptor (EGFR: cetuximab, panitumumab, dacomitinib) sporadic responses; checkpoint inhibitors ~1 in 6 respond — trials only.
- Chemoradiation (mitomycin + capecitabine + RT) for cisplatin-ineligible; International Penile Advanced Cancer Trial (InPACT) is testing sequencing.

13 · FOLLOW-UP

- Organ-preserving treatment: close follow-up and self-examination (recurrence higher; may be late after RT).
- Observed groins: years 1–2 every 2–3 months; year 3 every 4–6 months; year 4 six-monthly; then annually. Only ~⅓ of cN0 men who relapse in the groin survive 5 years → act on any change.
- Second primary neoplasms in 5–15%.

14 · NON-SQUAMOUS PENILE TUMOURS (~5%)

Tumour	Key features	Treatment
Basal cell carcinoma	Proximal shaft near base (sun-protected skin → rare); almost never metastasises. Fibroepithelioma of Pinkus = premalignant variant	Local excision / Mohs — curative
Melanoma	<1%; pigmented glans lesion; haematogenous (corpora) + lymphatic spread; staged by AJCC cutaneous system (Breslow, Clark)	Foreskin → circumcision; glans → partial penectomy; glans–shaft → partial/total. Bilateral modified ILND if Breslow ≥1 mm, ulceration or Clark IV–V; DSNB reasonable. Adjuvant nivolumab for resected stage IIIB–IV
Sarcoma	Malignant: **proximal** shaft (benign distal); vascular (haemangioendothelioma) commonest; leiomyosarcoma; may mimic Peyronie plaque or cause priapism. **Local recurrence characteristic**, metastasis rare	Superficial → wide excision / partial amputation; deep corporal → total; no node dissection unless palpable
Extramammary Paget disease	Eczematoid penoscrotal plaque (apocrine sites); Paget cells cytokeratin 7+, carcinoembryonic antigen+, S-100−; behaves as slow intraepithelial apocrine adenocarcinoma. ~½ have another cancer (prostate, bladder, kidney) → screen	Wide excision (up to 3 cm) guided by **mapping biopsies**; flap cover; ILND if nodes

Tumour	Key features	Treatment
Lymphoma	Penis has no lymphoid tissue → "primary" usually systemic; lump or priapism; diffuse large B-cell	Systemic rituximab, cyclophosphamide, doxorubicin, vincristine, prednisone (R-CHOP) even if isolated
Metastases	From **bladder, prostate, rectum**; **priapism** (corporal replacement) commonest sign; most die within a year (prostate origin longer)	Palliation: partial/total penectomy for pain or solitary distal disease, dorsal nerve section, suprapubic cystostomy; treat primary histology

15 · LANDMARK TRIALS

Trial	Design	Result	Take-home
Neoadjuvant TIP (MD Anderson)	Phase II; 4 cycles TIP for cN2–N3 M0	Response 50%; median overall survival 17.1 months; durable survival mainly in responders	Neoadjuvant TIP is standard for bulky/pelvic nodes before surgery
Netherlands Cancer Institute early vs delayed node surgery	Cohort: node-positive via early DSNB vs dissection after surveillance relapse	3-year survival 84% vs 35%; ENE nearly universal if delayed	Stage high-risk cN0 groins early

16 · MCQ ONE-LINERS

1. PeIN is defined by an intact basement membrane; WHO 2022 regards all PeIN as high grade.
2. Bowenoid papulosis: shaft of young men; does not progress to invasion.
3. Erythroplasia of Queyrat progresses in up to 33%; Bowen disease ~5%.
4. p16 immunohistochemistry is the surrogate marker for high-risk HPV.
5. HPV 16 is the commonest type in penile SCC; HPV is found in ~half.
6. Lichen sclerosus is linked to HPV-independent SCC.
7. Phimosis: 7–10-fold risk; adult circumcision gives little protection.
8. Neonatal circumcision protects against invasive cancer but not CIS.
9. Basaloid SCC: HPV-associated, >half node positive.
0. Verrucous carcinoma: pushing border, does not metastasise.
1. Sarcomatoid SCC is the most aggressive subtype.
2. Any proportion of anaplastic cells makes the tumour grade III.
3. pT1b = LVI, PNI, or high grade.
4. pT2 = corpus spongiosum; pT3 = corpora cavernosa.
5. Urethral invasion no longer upstages in TNM 8th edition.
6. pN1 = ≤2 unilateral inguinal nodes without ENE; pN3 = ENE or pelvic nodes.
17. Hypercalcaemia without bone metastases correlates with bulky inguinal disease.
18. No imaging reliably detects micrometastases in cN0 groins.
19. Negative surgical margin >1 mm suffices; the 2-cm rule is obsolete.
20. Nd:YAG preferred over CO_2 laser (6 mm vs 0.1 mm penetration).
21. Topical 5-FU/imiquimod is for Tis only.
22. Circumcision is mandatory before penile radiotherapy.
23. Brachytherapy suits T1–T2 glans tumours <4 cm.
24. Commonest late RT complication: meatal stenosis.
25. Most important prognostic factor: inguinal node status.
26. DSNB is the preferred staging for high-risk cN0 groins.
27. Antibiotic trial before FNAC of palpable nodes is no longer advised.
28. PLND if >2 positive inguinal nodes or ENE.
29. Bleomycin-containing regimens are discouraged for toxicity.
30. Penile BCC arises on the proximal shaft near the base.
31. Commonest sign of penile metastasis: priapism.
32. Commonest sources of penile metastases: bladder, prostate, rectum.
33. Penile lymphoma is treated systemically (R-CHOP).

84 Inguinal Node Dissection

Reading time ≈ 19 min (slow pace)

1 · WHY THE GROIN MATTERS IN PENILE CANCER

- Inguinal lymph node dissection (ILND) — first described by Basset (1912); Daseler (1948): radical ILND and five-zone anatomy [MCQ].
- Penile squamous cell carcinoma (SCC) has a **contained locoregional phase**: groin disease precedes systemic spread, so the groin is a **curable station**, not a marker of systemic disease.
- **Nodal status is the strongest prognostic factor**; extranodal extension (ENE), even in one node, = pN3.
- Early vs delayed ILND (impalpable T2–3 nodes): ENE ≈20% vs ≈95%; 3-year disease-specific survival ≈84% vs ≈35%. → Manage groins within **3 months** of diagnosis.
- Adequacy: ≥**8 nodes per side**. Guideline adherence is poor; omission predicts mortality — fear of morbidity must not cause omission.

2 · INDICATIONS AND RISK GROUPS

	Low risk	Intermediate	High risk
NCCN	pTis, pTa, pT1a	pT1b G1–2	≥pT2, or G3/G4
EAU-ASCO	pTa, pTis, G1 pT1a	G2 pT1a	≥pT1b
cN0 — NCCN	Surveillance	Modified/radical bilateral ILND or DSNB	Same
cN0 — EAU-ASCO	Surveillance	Surveillance (compliant) vs DSNB	DSNB; ILND if unavailable

NCCN = National Comprehensive Cancer Network; EAU-ASCO = European Association of Urology–American Society of Clinical Oncology; DSNB = dynamic sentinel node biopsy; G = grade. **Trap**: pT1b G2 is intermediate (NCCN) but high (EAU-ASCO).

- Staging tool, impalpable high-risk groin: **NCCN leads with ILND** (DSNB if experienced); **EAU-ASCO leads with DSNB** (ILND if DSNB unavailable or patient prefers). Low risk: surveillance without staging imaging.
- Selection uses **primary pathology only**: T stage, grade, lymphovascular and perineural invasion, sarcomatoid features. Human papillomavirus (HPV) status and nomograms are not decision tools.
- Expert pathology review: grading is poorly reproducible.
- **cN+ (palpable)**: ≈half are metastatic (rest reactive). Percutaneous fine-needle aspiration cytology (FNAC) → if negative, excisional biopsy → both negative: surveillance; any positive → **bilateral radical ILND**. Do not give an antibiotic trial (delays treatment).
- Unilateral palpable nodes → contralateral micrometastasis ≈**30%** → contralateral groin is staged.
- **Bulky/fixed (cN3)**: upfront cisplatin- and taxane-based chemotherapy, then surgery; upfront resection only if chemotherapy-ineligible or vessel invasion imminent. Skin involvement → wide excision + gracilis or rectus abdominis myocutaneous flap over vessels.

3 · STAGING WORK-UP

- Examine groins: cN0 vs cN+ (number, side, mobile/fixed). Indeterminate (obesity, prior surgery) → ultrasound (US), computed tomography (CT), magnetic resonance imaging (MRI) or positron emission tomography (PET)/CT.
- Palpable nodes/high risk → ^{18}FDG-PET/CT or CT chest–abdomen–pelvis for distant disease. Imaging misses deposits <1 cm → no role in the impalpable groin.
- **FNAC**: specific, insensitive → positive = ILND; negative = DSNB or ILND, never discharge.
- **DSNB**: US ± FNAC first → technetium-99m nanocolloid lymphoscintigraphy + blue dye. US is needed because a **tumour-replaced node blocks inflow**, rerouting tracer to a neo-sentinel node

→ false negative. Glans lesions may drain directly to deep nodes (another false negative). Serial sectioning reduces missed micrometastasis. Positive DSNB → ILND †; negative → surveillance.

KEY POINT Every groin test fails toward **false negatives**; a negative result never ends the pathway.

4 · LYMPHATIC ANATOMY

- Superficial system (prepuce, shaft skin) and deep system (glans, within Buck's fascia) → presymphyseal crossover → **bilateral drainage; side unpredictable** from lesion site.
- **Stepwise, no skip**: superficial inguinal → deep inguinal → external iliac. Pelvic spread requires ipsilateral inguinal disease; crossover to contralateral pelvis extremely rare.
- **Femoral triangle** (radical template): inguinal ligament above, **adductor longus** medially, **sartorius** laterally, apex where they cross.
- Superficial nodes (4–25) above fascia lata; deep nodes (1–3) medial to femoral vein in femoral canal. **Node of Cloquet**: most cephalad deep node, between femoral vein and lacunar ligament.
- Skin flaps raised deep to **Camper's fascia** along Scarpa's plane (capillary network in Camper's). Too thin → necrosis; too thick → nodes left behind. Incise parallel to Langer's lines.

Daseler zone	Related vessel
Central	Saphenofemoral junction
Superomedial (commonest sentinel)	Superficial external pudendal, superficial epigastric veins
Superolateral	Superficial circumflex iliac vein
Inferolateral	Lateral femoral cutaneous, superficial circumflex veins
Inferomedial	Great saphenous vein

Fossa ovalis order, lateral → medial: **nerve, artery, vein**. Femoral nerve lies lateral to the artery → **never dissect lateral to the femoral artery**.

5 · TEMPLATES

Template	Extent	Features
Radical	Whole femoral triangle, superficial + deep	Fascia lata opened; saphenous ligated; sartorius transposed
Modified (Catalona)	Excludes area lateral to femoral artery and caudal to fossa ovalis	Shorter incision, saphenous preserved, thicker flaps, no sartorius transposition
Superficial	Above fascia lata only	Fascia lata preserved

Modified ILND (cN0 staging) → frozen section; **positive → convert to radical ILND †**. Deep nodes are positive only when superficial nodes are positive, so negative superficial nodes allow omitting deep dissection.

6 · OPEN RADICAL ILND

ALGORITHM
Meticulous asepsis, frog-leg → oblique incision ≈2 cm below and parallel to inguinal ligament → flaps 3 cm above ligament to apex → sweep tissue off external oblique and cord → divide saphenous at apex (spare if minimal disease) → incise fascia lata over sartorius and adductor longus → dissect cephalad on anterior femoral vessels → divide saphenous at junction → clear deep nodes to femoral canal → **sartorius transposition** over vessels → close femoral canal (inguinal to Cooper's ligament) → suction drains, tack flaps.

- Large defects: scrotal rotation, abdominal advancement, or myocutaneous flaps; else negative-pressure therapy.

- Lymphatic control: vessel-sealing devices theoretically better than monopolar (which leaves lumens patent); fibrin sealant no benefit.

7 · PELVIC LYMPH NODE DISSECTION (PLND)

- Positive inguinal frozen section allows PLND at the same operation; otherwise staged. Ipsilateral PLND follows ipsilateral inguinal positivity.
- † Indications (EAU): ≥2 positive inguinal nodes or ENE (pN2–3) → ipsilateral PLND, by open or minimally invasive approach.

8 · VIDEO-ENDOSCOPIC ILND (VEIL)

- Video-endoscopic and robotic VEIL (RVEIL): same template, ports ≈25 cm below the ligament; work beneath Scarpa's fascia; insufflate 15 mm Hg; frozen section gates deep dissection; check haemostasis at 5 mm Hg; dependent drain.
- vs open: similar nodal yield and recurrence; shorter stay and drain time; fewer wound complications, flap necrosis and severe oedema. Only for **non-bulky, non-fixed** nodes; evidence retrospective.

9 · COMPLICATIONS, PREVENTION AND POST-OP CARE

Complication	Cause	Prevention / management
Wound infection, breakdown (commonest)	Skin flora, dead space	Asepsis, antibiotics; tack flaps; incisional negative pressure
Flap necrosis	Breach of Camper's capillary network	Correct plane, Langer-line incision, modified template, VEIL; debride + flap/negative pressure †
Lymphocele / seroma	Unsealed lymphatics	Sealing devices, suction drain until **<30 mL/day**; aspirate/drain symptomatic or infected collections †
Lymphoedema	Lymphatic interruption	Saphenous sparing, modified template, compression hose ≥1 month, lymphoedema referral
Others	Phlebitis, neurapraxia, pulmonary embolism, femoral vessel bleeding	Thromboprophylaxis; never dissect lateral to artery

Post-op: early mobilisation unless wound at risk; NCCN advises antibiotic prophylaxis (oral cephalosporin for days–weeks), the chapter authors do not routinely; prophylactic anticoagulation common. Obesity, smoking and poor healing raise risk.

Two root causes: disrupted **skin blood supply** (necrosis, dehiscence, infection) and disrupted **lymphatics** (lymphocele, lymphoedema). High-volume surgeons reduce both.

10 · FOLLOW-UP

Median inguinal recurrence ≈1 year; most within 2–3 years. EAU-ASCO: examination 3-monthly × 2 years, then 6-monthly to 5 years; imaging only if node-positive. NCCN images every patient after ILND (CT abdomen–pelvis + chest radiograph), including pN0. More intensive after DSNB (less complete staging). After 5 years, self-examination.

11 · MCQ ONE-LINERS

1. Strongest prognostic factor in penile cancer: nodal status.
2. Manage groins within 3 months of diagnosis.
3. Adequate ILND: ≥8 nodes per side.
4. ENE in a single node = pN3.
5. Commonest sentinel node zone: superomedial.

6. Femoral triangle: inguinal ligament, adductor longus medially, sartorius laterally.
7. Node of Cloquet: between femoral vein and lacunar ligament.
8. Modified ILND spares area lateral to femoral artery and below fossa ovalis.
9. Unilateral palpable node: contralateral micrometastasis ≈30%.
0. Penile lymphatic spread does not skip inguinal nodes.
1. Negative FNAC never ends the pathway.
2. Tumour-replaced node causes false-negative DSNB.
3. Sartorius transposition protects femoral vessels.
4. Flaps raised deep to Camper's fascia.
15. Commonest ILND complication: wound-healing problems.
16. Commonest complication after robotic ILND: lymphocele.
17. Fibrin sealant does not reduce lymphorrhoea.
18. cN3 nodes: neoadjuvant chemotherapy first.
19. Drain removal: output <30 mL/day.
20. Median time to inguinal recurrence ≈1 year.
21. Fossa ovalis, lateral to medial: nerve, artery, vein.
22. pT1b G2: intermediate risk (NCCN) but high risk (EAU-ASCO).
23. Glans tumours may drain directly to deep inguinal nodes.
24. VEIL suits only non-bulky, non-fixed nodes.

83 Tumours of the Urethra

Reading time ≈ 28 min (slow pace)

1 · BENIGN URETHRAL TUMOURS

Lesion	Who / where	Clue	Treatment
Leiomyoma	Reproductive-age women; posterior wall	Firm smooth suburothelial mass; grows in pregnancy (hormone sensitive)	Excision (transvaginal/transurethral); may recur locally
Haemangioma	Men, any age	Bluish sessile lesion, painless bleeding; posterior urethral → haematospermia; Klippel–Trénaunay–Weber association	Fulguration/laser; open excision if extensive; recurs if incompletely ablated
Fibroepithelial polyp	Boys; bulbar/posterior urethra	Pink polyp on a stalk; filling defect on urethrography	Transurethral resection; histology to exclude papilloma

Urethral prolapse and caruncle are mimics, not tumours.

2 · ANATOMY → HISTOLOGY → LYMPHATICS

Feature	Male (~20 cm)	Female (~4 cm)
Segments	Anterior: fossa navicularis, pendulous, bulbar. Posterior: membranous, prostatic	Anterior = distal ⅓; posterior = proximal ⅔
Epithelium	Prostatic: urothelium → anterior: columnar → fossa navicularis: squamous	Proximal ⅓ urothelium; distal ⅔ stratified squamous
Glands	Cowper's (bulbourethral), Littré	Skene's (paraurethral; prostate homologue)
Histology	Urothelial carcinoma (UC) 50–80%, squamous cell carcinoma (SCC) 10–30%, adenocarcinoma 5–10%. **Anterior: up to 80% SCC; posterior: mostly UC**; adenocarcinoma relatively more in bulbar	UC 28–45%, SCC 19–29%, **adenocarcinoma 28–38%** (near-equal)
Lymphatics (both sexes)	Anterior → **inguinal** (bulbar also external iliac); posterior → **pelvic**	

KEY POINT Site predicts histology (distal SCC, proximal UC), histology predicts behaviour, and the same anterior–posterior line fixes the nodal basin. In women the line is also the **continence boundary**: the distal third can be excised without incontinence.

- Grading is histology-specific: UC low/high grade only; SCC/adenocarcinoma G1–G3.

3 · STAGING — AMERICAN JOINT COMMITTEE ON CANCER (AJCC) 8TH EDITION

T	Penile urethra and female urethra	Prostatic urethra
Ta / Tis	Non-invasive papillary / carcinoma in situ (CIS)	Tis also includes prostatic duct involvement **without stromal invasion**
T1	Subepithelial connective tissue	Subepithelial connective tissue
T2	Corpus spongiosum or periurethral muscle	Prostatic stroma
T3	Corpus cavernosum or anterior vagina	Periprostatic fat
T4	Other adjacent organs (bladder)	Bladder, rectum
N	N1 single, N2 multiple regional nodes (inguinal, pelvic, presacral) — **number, not size**	

- **Evaluation**: examination under anaesthesia (EUA) with cystourethroscopy, bimanual palpation (genitalia, urethra, rectum/vagina, perineum, groins); transurethral or needle biopsy. Cytology poorly sensitive. **Magnetic resonance imaging (MRI)** for local extent; computed tomography (CT)/MRI abdomen–pelvis, CT chest if invasive.
- **Palpable inguinal nodes (20–30%) are almost always metastatic** — unlike penile cancer, not observed on antibiotics.

4 · MALE PRIMARY URETHRAL CANCER (PUC)

- Rare; >60 years. **Urethral stricture** is the commonest risk factor (up to 50%; latency 20–30 years). Others: sexually transmitted infection, lichen sclerosus, urethritis, pelvic radiotherapy (RT), trauma, instrumentation, human papillomavirus (HPV) 16.
- Present: obstructive voiding, haematuria/bloody discharge, penile or perineal mass, † periurethral abscess/fistula; nearly all symptomatic. † Biopsy a stricture that bleeds or recurs rapidly.
- **Prognosis**: stage and location — distal better than proximal (2–3-fold); bulbar survival as low as 20–30%. Proximal tumours present at higher stage.

Pendulous / fossa navicularis (distal)

- Usually resectable; surgery alone controls low-stage disease. Ladder: transurethral resection → **distal urethrectomy ± partial penectomy** → total urethrectomy + perineal urethrostomy ± total penectomy.
- **Penile preservation** endorsed (European Association of Urology, EAU; National Comprehensive Cancer Network, NCCN): topical 5-fluorouracil (5-FU) for distal CIS, distal urethrectomy + two-stage urethroplasty, glansectomy + graft, anterior urethrectomy + perineal urethrostomy. Positive margin → re-excision or adjuvant RT.
- **Margin: 5 mm** negative is now standard (1 cm traditional). Invasion reaching proximal penile urethra → total penectomy.
- Distal CIS can be upstaged → thorough staging, aggressive treatment.
- Penectomy controls nearly all pendulous tumours locally, yet systemic relapse is commoner than with fossa navicularis tumours — more proximal origin means greater systemic threat at equal local control.
- **Primary RT**: only for early anterior lesions in men refusing surgery (penis suspended on a catheter); preserves penis but causes stricture, ulceration, oedema. RT alone is ineffective for bulbar PUC.

Bulbar (proximal anterior)

- Few low-stage: transurethral resection or segmental excision + end-to-end anastomosis.
- **Advanced**: radical cystoprostatectomy + pelvic lymphadenectomy + total urethrectomy + penectomy ± inferior pubic rami and urogenital diaphragm (flap cover). Surgery alone → ~¼ 5-year survival → **multimodal therapy**.
- Bladder-sparing in selected: urethrectomy + perineal urethrostomy, or bladder-neck closure + catheterisable stoma.
- **Operative principles**: dorsal lithotomy; inverted-U perineal incision; develop ischiorectal fossae; mobilise corpora only a short way (to avoid breaching tumour) before inferior pubectomy; deliver penis through perineum; en bloc specimen.

Inguinal nodes (male and female)

- **No proven benefit of prophylactic inguinal dissection in cN0**, and no predictors to select patients. If primary irradiated and cN0 → consider prophylactic groin RT.
- cN1 → aggressive: neoadjuvant chemotherapy/chemoradiotherapy then lymphadenectomy (technique as for penile cancer). Nodal control also prevents skin breakdown and vascular erosion.

5 · MALE POSTERIOR (PROSTATIC) URETHRAL CANCER

- Usually **secondary to bladder UC**: 20–40% at radical cystectomy (RC); up to 39% of high-risk non-muscle-invasive bladder cancer (NMIBC) recurs in prostate after intravesical therapy. Primary prostatic urethral UC is rare → always evaluate bladder and upper tracts.
- Diagnosis by transurethral resection of the prostate (TURP) biopsy; often CIS. Historically all prostatic involvement was called T4 and treated by RC — many do well with less.
- BCG for high-grade superficial prostatic disease: ≥70% response; extensive TURP plus BCG gives ~half complete response, a minority progress to need RC.

Involvement	Treatment	5-year survival
Mucosal	TURP + bacillus Calmette–Guérin (BCG)	67–74%
Ductal / acinar (still Tis)	TURP + BCG, or RC ± urethrectomy	67–74%
Stromal invasion	**RC + urethrectomy + pelvic lymphadenectomy** ± neoadjuvant chemotherapy	~36%; follows bladder stage

> **KEY POINT** The only question is "stroma breached?" TURP before BCG debulks, improves BCG contact and stages. Ductal disease is not T2 — do not jump to cystectomy. Repeat prostatic biopsies on follow-up.

6 · FEMALE URETHRAL CANCER

- ~⅓ of PUC; >60 years. Risk: chronic irritation, recurrent infection, leukoplakia, caruncle, HPV, **urethral diverticulum (6–10% harbour malignancy**; mostly adenocarcinoma).
- Presents: haematuria/vaginal spotting, lower urinary tract symptoms, urethral/vaginal mass.
- **Histology by location**: distal → SCC; proximal → UC or adenocarcinoma. Adenocarcinoma: **mucinous** (commoner; carcinoembryonic antigen) and **clear cell** (diverticula; may secrete prostate-specific antigen, PSA); likely Skene's gland origin.
- **Work-up**: pelvic/bimanual exam (vaginal wall, vulva, groins), EUA + cystourethroscopy + biopsy, MRI pelvis, chest CT. **Location decides whether continence-sparing excision is possible**; ⅓ distal, rest proximal/pan-urethral.
- At presentation: 30–50% locally advanced; ~20% palpable inguinal nodes. **Stage is the dominant prognostic factor**; distal > proximal > pan-urethral. Female sex itself not adverse (later stage). Melanoma uniformly poor.

Treatment

Situation	Management
Small non-invasive	Endoscopic resection / laser ± BCG
Low-stage distal (anterior)	**Distal urethrectomy** (transvaginal, with anterior vaginal wall; frozen-section proximal margin) **or RT** (brachytherapy ± external beam, ~60 Gy) — equal options; ≥70% cured. Problems: stress incontinence (>40%), meatal stenosis (wide spatulation); ~1 in 5 recur locally. RT complications 20–50% (stricture, necrosis, fistula)
Proximal / pan-urethral	**Anterior exenteration**: RC + total urethrectomy + extended pelvic lymphadenectomy + anterior/whole vaginal wall ± hysterectomy, vulvectomy, pubic symphysis/rami (or intraoperative RT). Alone → most die, ~⅓ pelvic recurrence → **multimodal** (neoadjuvant chemotherapy or chemoradiotherapy then surgery/RT)
Palpable groin nodes	Chemoradiotherapy, chemotherapy, or lymphadenectomy; often curable. Pelvic nodes → poor

- Radical urethrectomy extent: periurethral tissue to bulbocavernosus laterally, pubic symphysis anteriorly, bladder neck superiorly, with anterior vaginal wall; perineal incision is inverted-U around the meatus.

- RT local control depends on **length of urethra involved**, not histology or size; combined external beam + brachytherapy beats external beam alone.

7 · MULTIMODAL THERAPY (BOTH SEXES)

- Surgery or RT alone fails in advanced/proximal disease → **cisplatin-based neoadjuvant chemotherapy** (EAU, NCCN) or **chemoradiotherapy**, then consolidative surgery; refer to high-volume centres. Response predicts survival.
- **Chemoradiotherapy**: 45–55 Gy to genitalia, perineum, groins + 5-FU and mitomycin C — preferred for advanced SCC; complete response common, similar proximal and distal; every responder develops stricture; ~⅓ recur locally. Not for low-stage distal disease (surgery).

Histology	Regimen
UC	Methotrexate, vinblastine, doxorubicin, cisplatin (MVAC); gemcitabine–cisplatin
UC or SCC	Paclitaxel, ifosfamide, cisplatin (TIP); cisplatin, gemcitabine, ifosfamide
Adenocarcinoma	Gemcitabine, 5-FU, leucovorin, cisplatin

Stage (NCCN, anterior male / female)	First line
Tis, Ta, T1	Repeat transurethral resection ± intraurethral BCG/chemotherapy
Male T2 pendulous	Partial urethrectomy ± partial penectomy; positive margin → chemoradiotherapy/surgery/RT
Male T2 bulbar	Urethrectomy ± RC; ≥pT3/N1 → adjuvant chemotherapy or chemoradiotherapy
Female T2	RC + urethrectomy, or distal urethrectomy by location, or chemoradiotherapy
≥T3 N0	Chemoradiotherapy ± surgery; or neoadjuvant chemotherapy → surgery/RT
N1	Chemoradiotherapy (preferred for SCC) or systemic therapy ± consolidative surgery

- Recurrence after resection of invasive disease ~50% → urethral, inguinal and systemic surveillance (lung, extrapelvic nodes). Metastatic disease: cisplatin chemotherapy ± resection can give long survival; programmed death-ligand 1 (PD-L1)-positive tumours may respond to checkpoint inhibitors.

8 · URETHRAL RECURRENCE (UR) AFTER RADICAL CYSTECTOMY

Male

- Risk **1–15%** (~5%); most within 2 years, ~11% after 5 years.
- **Risk factors**: **prostatic urethral involvement** (15% vs 5%), **positive urethral margin**, cutaneous diversion, CIS, multifocality, prior NMIBC. Orthotopic neobladder (ONB) protective (~5% vs 9%).
- Mechanism: direct spread along urothelium from the bladder primary, or a field defect (metachronous).
- Routine prophylactic urethrectomy abandoned (64–100% of specimens negative; added morbidity; rise of orthotopic diversion). Applying every risk factor would exclude ~70% from neobladder, so only the highest-risk men lose the urethra.
- **Before ONB**: negative intraoperative frozen section of the urethral margin (preferred over pre-op prostatic biopsy, which correlates poorly with the margin).
- **Urethrectomy** — absolute: positive margin or gross urethral tumour; relative: diffuse CIS, prostatic urethral involvement, multifocal, bladder neck or upper tract tumours. Staged later urethrectomy as good as simultaneous.

Female

- Synchronous urethral involvement 8–13%; main risk **bladder neck involvement** — but only 30–50% of those have urethral tumour → not an absolute contraindication.
- ONB if no bladder neck, urethral or anterior vaginal wall involvement and negative frozen section; otherwise concurrent urethrectomy. UR after ONB <5% — shorter urethra and post-menopausal cephalad squamous epithelium; most arise from skip lesions despite negative margins, so surveillance continues. Treatment: urethrectomy + cutaneous diversion.

Surveillance and treatment

- Presents with bloody discharge or mass. Urethral wash cytology (barbotage if cutaneous diversion) considered for 2 years — not shown to improve survival.
- Positive cytology/symptoms → urethroscopy + biopsy, restaging.
- **Total urethrectomy is treatment of choice** (entire urethra incl. meatus). After ONB → convert to cutaneous diversion (ileal conduit from neobladder bowel). Non-invasive: endoscopic resection; **intraurethral BCG for CIS** (not papillary/invasive).
- Isolated non-invasive UR: favourable; invasive UR median survival ~17 months; ≥⅓ have concurrent pelvic/distant recurrence. Endoscopic management carries a high second-UR risk.
- Urethrectomy after cutaneous diversion: the proximal (membranous) segment in the scarred pelvic floor risks a positive margin and retropubic relapse.
- **Male total urethrectomy steps**: lithotomy; perineal incision; mobilise spongiosum off corpora; invert penis to glans; excise meatus and glanular urethra; dissect bulb (ligate bulbar arteries posterolaterally); excise membranous remnant with frozen section; beware bowel on urogenital diaphragm; drain.

9 · MCQ ONE-LINERS

1. Commonest risk factor for male urethral cancer: urethral stricture.
2. Anterior male urethral cancer is mostly SCC; posterior mostly UC.
3. Female urethra: adenocarcinoma nearly as common as UC and SCC.
4. Urethral diverticulum carcinoma is usually adenocarcinoma.
5. Clear cell urethral adenocarcinoma may secrete PSA.
6. Palpable inguinal nodes in urethral cancer are almost always metastatic.
7. Prophylactic inguinal dissection has no proven benefit in cN0 urethral cancer.
8. Prostatic stromal invasion = T2 prostatic urethra.
9. Prostatic ductal involvement without stroma is Tis.
0. MRI is optimal for local staging of urethral cancer.
1. Distal male urethrectomy margin of 5 mm is acceptable.
12. Distal female urethra can be excised without incontinence.
13. Female urethral RT local control depends on length of urethra involved.
14. Chemoradiotherapy (5-FU + mitomycin C) preferred for advanced squamous PUC.
15. Every responder to urethral chemoradiotherapy develops stricture.
16. MVAC is active only against urothelial histology.
17. Male urethral recurrence after cystectomy is ~5%.
18. Prostatic urethral involvement triples male urethral recurrence risk.
19. Orthotopic neobladder lowers urethral recurrence versus cutaneous diversion.
20. Positive urethral margin: absolute indication for urethrectomy.
21. Female cystectomy: bladder neck involvement is main urethral risk.
22. Intraurethral BCG works for urethral CIS, not papillary disease.

85 Benign Disorders of the Penis and Urethra

Reading time ≈ 28 min (slow pace)

1 · RECONSTRUCTIVE SURGICAL TECHNIQUE

- Reconstruction succeeds or fails on the wound bed, so every choice preserves blood supply and limits trauma.
- **Cautery**: **bipolar on the urethra** (both electrodes in the handpiece → minimal mucosal/spongiosal damage); monopolar on subcutaneous tissue and muscle.
- **Bougie à boule** = calibration (essential). **Goodwin sounds** = dilation, safer than van Buren (lumen for guidewire); either can pass via the suprapubic tract to the bladder neck in posterior urethroplasty.
- **Sutures**: always **absorbable** (permanent suture = stone nidus); monofilament polydioxanone (PDS) preferred. Wound support needed **30–60 days**; chromic lasts only 14 days → avoid. **5-0** for most urethral work; **6-0** for single-layer mucosal anastomosis (non-transecting excision and primary anastomosis, EPA); small taper needle (RB-1).
- **Position**: limit high lithotomy (neuropathy, compartment syndrome); low lithotomy during graft harvest.

2 · TISSUE TRANSFER: GRAFTS AND FLAPS

- **Graft** = separated from its blood supply, must acquire a new one from the host bed. **Flap** = transferred with its blood supply.

Graft take (~96 h)

Phase	Duration	Mechanism
Imbibition	~48 h	Graft "drinks" nutrients from the bed; temperature below core
Inosculation	~48 h	True microcirculation re-established; temperature rises to core

Anything impairing host-bed vascularity impairs take. Immobilise skin grafts **5–10 days** (bolster or vacuum over petroleum gauze).

Split- versus full-thickness skin graft

Feature	Split-thickness (STSG)	Full-thickness (FTSG)
Layers	Epidermis + part of dermis (no reticular dermis)	Epidermis + full dermis; subdermal plexus exposed
Plexus / take	Many small vessels → reliable take	Large sparse vessels → "fastidious"
Contraction, durability	Contracts; brittle	Contracts less; durable
Hair, size	No adnexa; large	Carries hair; size limited

- Skin in contact with urine must be **hairless** (pre-harvest laser/electrolysis).
- **Extragenital FTSG** fails in the urethra (greater mass → ischaemic take). Exception: **posterior auricular (Wolfe) graft** — thin skin on perforators, behaves like an STSG.

Other grafts: dermal and vein (tunical defects in Peyronie disease, PD); tunica vaginalis (easy harvest); bladder epithelium and rectal mucosa (urethra when oral mucosa unavailable).

Oral mucosa graft (OMG)

- Non-keratinised, elastin-rich epithelium on a vascular lamina propria with a **panlaminar plexus**; used to wet environment; holds sutures, sacculates less than skin.
- **Buccal mucosa graft (BMG)**: more early swelling and numbness; **lingual**: speech difficulty; labial: worst morbidity. Buccal = lingual in success.

- Buccal onlays are durable; failures occur within the **first 12 months**.

> **KEY POINT** Graft trade-off: dense fine plexus and low mass favour take; carried dermis/lamina gives durability and resists contraction. This explains STSG vs FTSG, failure of extragenital FTSG, the Wolfe exception, and why OMG is ideal.

Flaps

Classified on independent axes, so one flap sits in several categories:

- **Composition**: fasciocutaneous (distal penile circular), fascial (dartos), musculocutaneous (rectus abdominis), muscle (gracilis), visceral (omentum).
- **Blood supply**: axial (named vessel in long axis — gracilis, penile circular flap) vs random (U-flap, **Z-plasty**). Axial peninsula (skin intact at base), island (skin divided, pedicle intact), free (microvascular).
- **Movement**: advancement (Y-V), rotation (Martius), transposition (Z-plasty). **Location**: local, distant pedicled, free (radial forearm phalloplasty).

3 · SURGICAL ANATOMY (ESSENTIALS)

- Layers, deep → superficial: **tunica albuginea → Buck's fascia → dartos → skin**. Buck's encloses deep dorsal vein, dorsal arteries and nerves and splits around the spongiosum; dartos is fat-free areolar tissue carrying the skin's vessels and fuses with Colles fascia.
- Urethra: **eccentric (dorsal) in the bulb**, central in the shaft; fossa navicularis stratified squamous; membranous urethra surrounded by the striated external sphincter.
- **Skin supply**: superficial external pudendal (femoral) → subdermal plexus (McAninch circular flap); posterior scrotal artery → ventral longitudinal flap.
- **Deep supply**: internal pudendal → common penile artery → **bulbourethral, dorsal (circumflex branches to spongiosum, arborises in glans), cavernosal**.

Situation	Spread
Buck's intact (penile fracture)	Confined to shaft → **eggplant deformity**
Buck's ruptured (straddle)	Within Colles–Scarpa: scrotum, perineum, abdominal wall; limited on thigh by fascia lata
Fournier gangrene	Perineum/scrotum → abdominal wall via Scarpa

> **KEY POINT** Two envelopes, two vascular worlds: dartos carries skin supply (so skin flaps are raised on dartos); Buck's encloses the erectile bodies fed by the common penile artery.

4 · LICHEN SCLEROSUS (LS)

- Formerly balanitis xerotica obliterans (BXO): chronic inflammatory genital skin disease of prepuce, glans, urethra; male peak 30–50 years.
- Pale, sclerotic, hypopigmented plaques; phimosis; **meatus most commonly affected** (sentinel of later stricture); pain, pruritus, obstructive voiding; chronic, progressive.
- Diagnosis clinical; biopsy if unclear (band-like lymphocytic infiltrate, epidermal atrophy).
- Cause unclear (autoimmune, genetic proposed); **chronic urine irritation** (dribbling, obesity) certainly increases risk.
- Stricture mechanism: pressure voiding behind meatal stenosis → urine intravasation into **glands of Littre** → inflammation → spongiofibrosis. **LS is limited to the anterior urethra.**
- Malignant transformation up to **8%** (mostly squamous cell carcinoma, SCC) → possibly premalignant.

Management

- **Topical steroid**: clobetasol propionate or betamethasone 0.05% once–twice daily 1–2 months, then alternate days 3–4 months.
- Poor surgical candidates → self-dilation lubricated with clobetasol.
- Meatal stenosis → **retrograde urethrography (RUG) first** (proximal stricture often coexists) → extended meatotomy (inverted-V relaxing incision prevents puckering) or BMG onlay; restenosis commoner than non-LS.
- Phimosis only → circumcision. Early meatal/fossa repair may avert panurethral stricture. Use **oral mucosa, never genital skin**; staged or single-stage. Recurrence higher than non-LS.
- Panurethral/buried disease → **perineal urethrostomy** (easy: proximal urethra spared) or staged reconstruction (stricture first, skin later).

> **KEY POINT** LS is a progressive skin disease sparing the proximal urethra: hence meatus as sentinel, RUG before meatotomy, oral mucosa not skin, high recurrence, and easy perineal urethrostomy.

5 · OTHER URETHRAL LESIONS

Lesion	Key features	Management
Urethral haemangioma	Haematuria, bloody discharge; meatus common; extends beyond cystoscopic view → **magnetic resonance imaging (MRI)**; always benign	Asymptomatic → observe; small → laser or sclerosant (triamcinolone); large → excision + staged reconstruction (avoid tubed grafts)
Reactive arthritis (Reiter)	Arthritis, conjunctivitis, urethritis after Chlamydia/Shigella/Salmonella; human leucocyte antigen (HLA)-B27 ~2/3; **circinate balanitis**	Urethritis mild, self-limiting
Amyloidosis	Rare urethral mass, obstruction, haematuria; cystoscopy + biopsy	Observe; dilation or urethroplasty if obstructive
Urethrocutaneous fistula	Mostly iatrogenic (hypospadias repair, urethroplasty in phalloplasty); piercing, periurethral infection behind stricture	Treat the **underlying process** (e.g. distal stricture) as well as the tract
Male urethral diverticulum	Congenital (all wall layers) vs acquired (no smooth muscle; behind stricture, catheter, trauma); dribbling, urinary tract infection (UTI), stone	Cystourethroscopy + RUG; repair with treatment of cause
Enlarged utricle / prostatic diverticulum	Müllerian remnant, proximal hypospadias; missed on voiding study	Catheter may enter utricle (false drainage); recurrent UTI, post-void "incontinence"

6 · PREPUCE AND MEATUS

Paraphimosis — emergency

- Retracted foreskin trapped behind corona → venous congestion → oedema → glans necrosis; classically after catheterisation without replacing foreskin.
- **Reduction**: steady compression for minutes to express oedema; thumbs push glans while fingers pull foreskin forward; never sudden force (granulated sugar osmotic aid suggested).
- Irreducible → emergency **dorsal slit** or circumcision. Recurrence common → elective dorsal slit/circumcision.

Phimosis and balanitis

- Phimosis: physiological in infants (resolves 1–8 years); in adults → balanitis, penile cancer. Topical steroid, dorsal slit, circumcision.
- Balanitis: **Candida** commonest; also sexually transmitted infection (STI), chemical, Zoon (plasma cell). Keep clean/dry, antifungal, treat STI, **check blood glucose** (diabetes). Recurrent balanitis

raises cancer risk.

Meatal stenosis (adult)

- After instrumentation/catheter, LS, hypospadias repair (child: post-circumcision → *Ch 55*).
- Simple → **ventral meatotomy** with sutures at apex and each side; or steroid-lubricated self-dilation (LS). Extension into fossa → urethroplasty → *Ch 86*.

Adult circumcision (sleeve technique)

ALGORITHM
Penile block at base → foreskin retracted: mark cuff incision, straight across frenular base → incise to Buck's → foreskin advanced: mark coronal incision ± frenuloplasty → excise sleeve with **bipolar** haemostasis (adult vessels do not seal by pressure) → quadrant dartos sutures → skin closure

- Too much skin removed → painful erection, buried penis; too little → adhesions, skin bridges. Transient glans hyperaesthesia.
- **Human immunodeficiency virus (HIV)**: foreskin rich in CD4+ cells, anaerobic microbiome, microtears; circumcision ↓ female-to-male HIV ~**60%**; also ↓ herpes simplex, human papillomavirus. Men who have sex with men: benefit mainly for insertive partners.

7 · PENILE CURVATURE

Curvature = asymmetric tunical compliance or shortening of one aspect.

Devine–Horton type	Abnormal tissue
I	Spongiosum deficient at curvature (chordee without hypospadias, I–III)
II	Dysgenetic Buck's + dartos band; urethra/spongiosum normal
III	Dartos only
IV	All normal; asymmetric corporal tunica = **congenital curvature**
V	Congenital short urethra — so rare its existence is doubted

- **Chordee without hypospadias** (forme fruste of hypospadias): ventral ± torsion; erection commensurate with flaccid size; hooded prepuce, ventral inelasticity.
- **Congenital curvature** (type IV): young men 18–30, ventral or lateral (left), smooth whole-shaft curve; **large erection**, often above-average length; painless.
- **Rule**: if the meatus reaches the glans tip, **do not divide the urethra** — the ventral corpora, not spongiosum, usually limit straightening.

ALGORITHM
Excise ventral dysgenetic tissue + mobilise spongiosum glans → perineum → artificial erection → ventral midline septotomy → still curved: dorsal tunical ellipse excision/plication under elevated neurovascular bundle (dorsal arteries may be dominant glans supply)

Acquired non-Peyronie curvature

- Peyronie disease → *Ch 76*; deep urethrotomy into tunica; neglected fracture; **subclinical fracture** (buckling without detumescence → lateral scar, indentation); septal insertion tear (dorsal, mimics PD).
- Unlike PD: normal erections, no veno-occlusive dysfunction, no shortening. Treat by **scar excision + graft**, not contralateral plication (second scar, near-circumferential indentation).

8 · PENILE TRANSPLANTATION

Vascularised composite allotransplantation after devastating loss: natural appearance, erection, normal voiding — but lifelong immunosuppression and psychological/ethical issues. Alternative: phalloplasty.

9 · ADULT ACQUIRED BURIED PENIS (AABP)

- Risks: **obesity**, aggressive circumcision, scrotal lymphoedema, LS.
- **Vicious cycle**: escutcheon → penis tethered by suspensory ligament → telescoping → trapped urine → inflammation, LS, scarring → more trapping. Hence **weight loss alone rarely cures**.
- **Pariser** classification by reconstructive need: I skin flap, II graft, III scrotoplasty, IV escutcheonectomy, V panniculectomy.
- LS/phimotic cicatrix alone → trial topical steroid; otherwise surgical.

Work-up

- Examine supine and standing; computed tomography (CT) if hernia suspected before panniculectomy.
- **Evaluate urethra** (high stricture rate): severe penile stricture → urethroplasty **before** grafting. Optimise diabetes; consider weight-loss referral; set expectations.

ALGORITHM
Dorsal/ventral slit of phimotic ring → glans traction → circumcising incision, excise non-viable skin (keep shaft-base skin) → escutcheonectomy (harvest graft from it first; keep 2–3 cm skin at base) → tack Scarpa to rectus fascia/pubis → drains → cover shaft with STSG (0.012–0.020 inch), fibrin glue, bolster 5–7 days

- Dressing off ~day 5; drains out <30 mL/day.

Scrotal lymphoedema

- Penis buried in massive scrotum. Causes: congenital (Milroy — infancy; Meige — puberty; tarda — adult), infective (**Wuchereria bancrofti**, lymphogranuloma venereum), hidradenitis, malignancy, node dissection, radiation.
- Resect **all** involved tissue; cover with spared perineal/lateral scrotal skin; protect and tack elongated cords; vessel-sealing device.

Foreign body injection

- Silicone, paraffin, petroleum jelly → delayed fibrosis, deformity, necrosis, abscess, Fournier gangrene.
- Infection → antibiotics, drainage, debridement; otherwise excise all affected tissue + STSG. Screen for **body dysmorphic disorder**.

10 · LANDMARK TRIALS

Trial	Design	Result	Take-home
African circumcision trials (South Africa, Kenya, Uganda)	Three RCTs, adult male circumcision vs none	~60% ↓ female-to-male HIV acquisition	Circumcision is an HIV-prevention strategy

11 · MCQ ONE-LINERS

1. Graft take: imbibition (~48 h) then inosculation (~48 h).
2. STSG: reliable take but contracts; FTSG: fastidious but durable.
3. Posterior auricular (Wolfe) FTSG behaves like an STSG.
4. OMG advantage: panlaminar plexus, thick lamina propria, wet-environment tolerance.
5. Z-plasty is a random transposition flap.
6. Bulbar urethra lies eccentric (dorsal) in the spongiosum.
7. McAninch flap relies on the superficial external pudendal subdermal plexus.
8. Commonest site of male genital LS: meatus.
9. LS never involves the posterior urethra.
10. LS malignant transformation: up to 8%, mainly SCC.
11. LS meatal stenosis: RUG before meatotomy.
12. Never use genital skin for LS urethroplasty.
13. Circinate balanitis: reactive arthritis.
14. Balanitis: commonest organism Candida.
15. Paraphimosis irreducible: emergency dorsal slit.
16. Circumcision reduces female-to-male HIV by ~60%.

7. Devine–Horton type IV = congenital curvature from tunical asymmetry.
8. Chordee without hypospadias: erection commensurate with flaccid size.
9. Meatus at glans tip: never divide urethra for ventral curvature.
20. Subclinical fracture curvature: excise and graft, not plicate.
21. Main risk factor for adult buried penis: obesity.
22. Classic infective cause of scrotal lymphoedema: Wuchereria bancrofti.

86 Urethral Stricture Disease

Reading time ≈ 29 min (slow pace)

1 · DEFINITIONS AND TERMINOLOGY

- **Urethral stricture** = fixed anatomic narrowing such that the lumen will not accept instrumentation without disrupting the mucosa. The term is reserved for the **anterior (spongiosum-invested) urethra**.
- Scar may extend through the spongy erectile tissue = **spongiofibrosis**; its contraction narrows the lumen.
- Area falls with the square of diameter: **halving calibre removes ~¾ of the lumen** — so contraction is silent, then symptoms appear abruptly.

Site	Correct term
Anterior urethra	Stricture (epithelial scar ± spongiofibrosis)
Membranous urethra after pelvic fracture	Distraction defect; pelvic fracture urethral injury (PFUI)
Posterior urethra after trauma/radical prostatectomy	Posterior urethral stenosis
Prostatic urethra, bladder neck	Contracture or stenosis

Jordan spectrum: mucosal fold → iris constriction → full-thickness ± spongiofibrosis → fibrosis beyond spongiosum → complex stricture with fistula.

2 · SURGICAL ANATOMY RELEVANT TO STRICTURE

- **Bulbar urethra is eccentric — close to the dorsum** of the spongiosum (basis of dorsal grafting and of the danger of deep 12 o'clock urethrotomy); penile urethra central.
- Corpus spongiosum (from the common penile artery, terminal internal pudendal) has **dual supply**: antegrade from the bulbar arteries and **retrograde** via dorsal arteries arborising in the glans.

> **KEY POINT** Retrograde glanular flow is what allows the bulb to be transected and mobilised. Where it is lost (hypospadias, common penile artery injury), mobilisation becomes ischaemia — the thread linking vessel-sparing excision and PFUI repair failure.

3 · AETIOLOGY

Any injury to epithelium or spongiosum that heals with scar. Commonest: **idiopathic ≈ iatrogenic** > traumatic > inflammatory.

Cause	Mechanism / points
Iatrogenic	Cystoscopy, dilation, transurethral resection of bladder tumour (TURBT) and of prostate (TURP) — mucosal insult, ischaemia (calibre, duration); catheters — pressure necrosis, infection (silicone/hydrophilic catheters may reduce)
Traumatic	**Straddle**: bulb crushed against inferior pubic rami → often unrecognised until obstruction; pelvic fracture → posterior distraction; prostate cancer treatment → posterior stenosis
Inflammatory	Gonorrhoea (historic, now rare); **lichen sclerosus (LS)**. No established link with Chlamydia/Ureaplasma
Idiopathic	No identified cause
Congenital	Rarest: short, non-inflammatory anterior stricture in infants before ambulation (no trauma potential)

- **LS**: starts as glans inflammation → meatal stenosis → fossa stricture. Proposed: high-pressure voiding → urine intravasation into **glands of Littre** → microabscesses → deep spongiofibrosis. Antibiotics relieve symptoms, not stricture.

> **KEY POINT** Aetiology predicts biology: iatrogenic and straddle strictures are focal ischaemic/crush scars suited to excision; LS is a progressive skin disease that recurs in any skin used to repair it.

4 · PRESENTATION AND EVALUATION

- Obstructive voiding, urinary tract infection, retention (after long-tolerated symptoms).
- **Acute retention**: failed catheter → retrograde urethrogram (RUG); rather than blind dilation, place a **suprapubic cystostomy** to buy planning time. Blind filiforms/dilation condemned; use flexible endoscopy and pass a guidewire under vision.
- Four determinants: **location, length, depth, density** of spongiofibrosis.

Modality	Contribution	Caveat
Dynamic RUG + voiding cystourethrogram (VCUG) (McCallum criteria)	Location, length; steep oblique estimates length	Dynamic, not static; done by/with treating surgeon
Sonourethrography	Length of narrow segment	May underestimate spongiofibrosis
Flexible urethroscopy	Appearance, compliance; paediatric scope sees beyond narrowing	Do not dilate at first evaluation
Magnetic resonance imaging (MRI)	Tumours; alignment in PFUI	Not for routine strictures

- **Hydraulic trap**: voiding pressure holds proximal diseased urethra open; once the narrow segment is relieved it contracts. Coning down = scarred; normal compliance = uninvolved.
- **Urethral rest**: suprapubic tube for **6–8 weeks** lets the hydrodilated urethra contract → true length.
- At surgery calibrate the whole urethra with bougie à boule.

5 · DECISION MAKING

- First decide **palliative vs curative** with the patient; periodic dilation is acceptable if that is the agreed goal.
- The **reconstructive ladder is archaic**: go to the procedure most likely to cure.

ALGORITHM

Short (<1.5 cm) bulbar, no dense spongiofibrosis, first treatment → dilation or internal urethrotomy (once) → recurrence → urethroplasty
Bulbar 1–2 cm (up to 3–5 cm proximally) → excision and primary anastomosis (EPA)
Longer bulbar → oral mucosa graft (OMG) onlay ± augmented anastomosis
Penile → onlay (graft or flap); hostile bed/LS/failed hypospadias → staged OMG
Panurethral/exhausted options/elderly → perineal urethrostomy

6 · DILATION

- Curative only for an **epithelial stricture without spongiofibrosis**. Goal: stretch scar without new scar — **bleeding means tearing**.
- Safest: **balloon dilation** over a guidewire placed endoscopically.
- **Optilume**: paclitaxel-coated balloon (inhibits fibroblasts) — see trials.

7 · INTERNAL URETHROTOMY (DVIU)

- Direct vision internal urethrotomy (DVIU): incision through scar to healthy tissue so the lumen heals larger, by **secondary intention** — a race between epithelialisation and wound contraction; recurrence if contraction wins. Worse with greater luminal narrowing (**>74%**).

- 12 o'clock cut questionable: spongiosum thinnest 10–2 o'clock; deep cuts enter intracrural space or corpora → erectile dysfunction (ED). Deep spongiofibrosis cannot be cured endoscopically, so deep cuts are pointless.
- Complications: **recurrence** (commonest), bleeding with erections, extravasation (use saline), spongiosum–cavernosum fistula.
- No success in non-bulbar strictures; **a second urethrotomy adds virtually nothing**; repeated procedures worsen later urethroplasty.

Favourable	Unfavourable
Bulbar	Penile, fossa
<1.5 cm	Longer
No dense deep spongiofibrosis	Dense (e.g. straddle)
First procedure	Repeat

Adjuncts against contraction

- Prolonged catheter (6 weeks) no better than 3–7 days. **Self-catheterisation** tapering over 3–6 months: recurs when stopped; valid palliation.
- **Mitomycin C** injection decreases recurrence. Laser = cold knife. UroLume permanent stent withdrawn.

> **KEY POINT** No endoscopic adjunct removes spongiofibrosis, so endoscopy cures only the short superficial bulbar stricture; repetition spends the patient's reconstructive capital.

8 · EXCISION AND PRIMARY ANASTOMOSIS (EPA)

- **Gold standard**, most durable (recurrence rare, mostly within 6 months). Three rules: **excise all fibrosis; wide spatulation (large ovoid anastomosis); tension-free**.
- Length gained by: vigorous spongiosal mobilisation, freeing Buck's fascia, developing the intracrural space, detaching bulbospongiosus from perineal body (saves ~1–1.5 cm).
- 1–2 cm easily; 3–5 cm in some. **The more proximal, the longer the excisable stricture**; younger tissue more compliant.
- Absorbable sutures: 3-0/4-0 for anastomosis, 4-0 to 6-0 for grafts/flaps; bipolar cautery near spongiosum.

	Transecting	Vessel-sparing transecting	Non-transecting
Bulbar arteries	Divided	Preserved	Preserved
Steps	Excise, spatulate, anastomose	Divide at membranous–bulbar junction, dorsal spatulation	Dorsal stricturotomy → excise scar from within → close transversely
Use	Most	Proximal bulbar	Proximal

Vessel-sparing protects men with poor retrograde supply (hypospadias, trauma, prior surgery); theoretically less ED and less artificial sphincter cuff erosion; advantage unproven.

9 · SUBSTITUTION URETHROPLASTY

Grafts

- **Buccal mucosa graft (BMG)** — workhorse for one-stage onlay and staged repair; full-thickness skin graft (FTSG, avoid in LS); meshed split-thickness skin graft (STSG) staged only; posterior auricular (Wolfe) graft staged hypospadias.
- Best in the **bulb**: thick, vascular spongiosum is a reliable bed.

Position	Features
Ventral	Only if spongioplasty planned

Position	Features
Lateral	Thin lateral spongiosum → less bleeding
Dorsal (Barbagli)	Dorsal urethrotomy → graft spread-fixed on corpora → edges sutured to graft

- **Augmented anastomosis**: excise the obliterated segment, floor (or roof) strip anastomosis, onlay graft over the remaining stricturotomy — often better than pure onlay.
- **Kulkarni**: one-stage panurethral dorsal onlay via perineal incision and **penile invagination**.

Staged repair

- **Stage 1**: stricture laid open; mesh STSG, BMG or Wolfe graft quilted to dartos (not on tunica — cannot be tubed). **Stage 2**: tubularise (Thiersch) to ~28 Fr after ≥6 months (12 after STSG).
- Indications: LS, complicated failed hypospadias, the most difficult cases.

> **KEY POINT** A graft lives on its bed — it succeeds where the bed is thick and vascular (bulb, dorsal corpora) and is staged when the bed, or skin disease, cannot be trusted in one step.

Flaps

- Genital skin islands on penile dartos or scrotal tunica dartos; **Quartey**: the skin is a passenger on a fascial flap. Skin must be hairless.
- Designs: ventral longitudinal (**Orandi**); dorsal transverse; hockey stick (7–9 cm); **circular island** — up to **15 cm**; scrotal island (not first choice).
- **Fossa navicularis**: dartos-based transverse (Jordan) or longitudinal (De Sy) island → excellent function and cosmesis.
- Grafts = flaps (meta-analysis); grafts easier; flaps for irradiated or multiply operated urethra.
- **Onlay beats tube**; tubed flap beats tubed graft. Very long strictures: flap distally + graft proximally.

Lichen sclerosus strictures

- Skin flaps and skin grafts failed (skin becomes diseased) → **oral mucosa only**, often staged; perineal urethrostomy for panurethral disease.

Perineal urethrostomy (PU)

- For failed repairs, LS, failed hypospadias, elderly/high-risk. **BMG PU**: inverted-U incision → spongiosum transected → dorsal urethrotomy to healthy mucosa → healthy ventral urethra to skin, calibrate 30 Fr → BMG quilted to corpora completes the dorsal wall (prevents circumferential scar).

10 · MEATAL STENOSIS

- **Child**: post-circumcision ammoniacal meatitis → ventral membrane → ventral meatotomy + parental dilation 3×/day for 7–10 days; early meatitis → 0.05% clobetasol.
- **Adult**: rarely isolated — fossa usually involved → evaluate as stricture. Ventral meatotomy with three sutures; dorsal Y-V meatotomy if needed.

11 · PELVIC FRACTURE URETHRAL INJURY (PFUI)

- In ~**10%** of pelvic fractures. Commonest site: **bulbomembranous junction**, not prostatic apex ("apple off its stem" is wrong). Prepubertal: may extend into prostatic urethra.
- Many are incomplete. **Primary realignment** (endoscopic catheter) may allow healing or ease later repair; controversial (linked to delayed repair, more endoscopic failures). Follow closely after catheter removal.

Evaluation

- Simultaneous **cystogram + RUG** defines gap if posterior urethra fills; else **antegrade flexible endoscopy via suprapubic tract + RUG**. Normal anterior urethra on RUG predicts successful anastomosis.

- Bladder neck appearance **does not predict** function — restore continuity first, treat incontinence later. MRI in difficult cases (high prostate → infrapubectomy likely).

Repair — delayed perineal anastomosis

- Timing: **3–6 months** after trauma. Goal = primary anastomosis in all until proven impossible. Transpubic/abdominoperineal unnecessary; **endoscopic "core-through" not curative**.

ALGORITHM
On-table endoscopy (exclude bladder stones) → exaggerated lithotomy → perineal incision → bulbospongiosus divided, bulb mobilised → Haygrove sound via suprapubic tract to proximal stump → excise fibrosis → spatulate both ends to **≥30 Fr** → 8–10 tension-free 4-0 polydioxanone (PDS) sutures over silicone stent

Progressive perineal approach (each step only if tension persists):

1. Distal urethral mobilisation (to penoscrotal junction, not beyond — chordee).
2. Develop intracrural (corporal) separation.
3. Inferior pubectomy (infrapubectomy, Kerrison rongeur).
4. Corporal rerouting of the urethra lateral to one crus.

- Postoperatively: suprapubic diversion, urethral stent; anticholinergic. **Contrast voiding trial at 21–28 days**; flexible endoscopy at ~6 months.
- Success: high **90s %**. Failure is mostly **ischaemic stenosis** of the mobilised bulb (now a flap on retrograde flow), not anastomotic.

Internal pudendal status	Potency	Action
Intact ≥1 side	Often potent	Reconstruct
Reconstituted only	Never potent	Reconstruct; revascularisation for ED later
Bilateral occlusion	—	**Revascularise first** (ischaemic stenosis risk)

Screen at-risk men (dorsal nerve injury, failed repair elsewhere, lateral-impact fracture) by **duplex ultrasound → angiography if abnormal**.

12 · VESICOURETHRAL ANASTOMOTIC STENOSIS AND FISTULA

- Endoscopic: deep 3 and 9 o'clock incisions; radial incision + mitomycin C for recalcitrant stenosis. Obliterated → robotic redo anastomosis or perineal repair (high incontinence; later artificial urinary sphincter, AUS). Suprapubic tube or diversion always options.
- **Rectourethral fistula**: simple (<2 cm, non-irradiated) → perineal or York-Mason; complex (irradiated, infected) → faecal diversion first, transperineal repair with **gracilis interposition ± BMG**; diversion often the honest endpoint after radiation.

13 · COMPLICATIONS AND FOLLOW-UP

- Recurrence (EPA early; grafts within 12 months; tissue transfer deteriorates with time); ED ≈ circumcision rate, higher with longer repairs and straddle, often improves; chordee; positioning neuropathy (limit lithotomy).
- Follow-up: symptoms, †uroflowmetry and post-void residual, urethroscopy if suspicious; recurrence → revision or periodic dilation.

14 · LANDMARK TRIALS

Trial	Design	Result	Take-home
ROBUST III	Phase III randomised controlled trial (RCT), Optilume vs endoscopic control after multiple failed endoscopic treatments	1-year stricture-free 74.6% vs 26.8%	Drug-coated balloon beats repeat endoscopy; role as first-line undefined

15 · MCQ ONE-LINERS

1. "Stricture" applies only to the anterior urethra.
2. Halving calibre cuts area ~75%.
3. Commonest causes: idiopathic, iatrogenic.
4. Bulbar urethra is dorsally eccentric in the spongiosum.
5. LS stricture mechanism: urine intravasation into glands of Littre.
6. Length assessment: dynamic RUG + VCUG.
7. Urethral rest for 6–8 weeks reveals true stricture length.
8. Bleeding during dilation means the scar is torn.
9. DVIU favourable: bulbar, <1.5 cm, first attempt.
0. Second urethrotomy adds virtually nothing.
1. EPA rules: total excision, spatulation, no tension.
2. Barbagli: dorsal graft spread-fixed onto corpora.
3. Grafts and flaps give equivalent results.
4. Onlay outperforms tubularised repair.
5. Avoid genital skin in LS strictures.
6. PFUI site: bulbomembranous junction.
7. PFUI repair timing: 3–6 months.
8. Endoscopic core-through is not curative for PFUI.
9. Progressive approach order: mobilisation, crural separation, infrapubectomy, rerouting.
0. PFUI repair failure: ischaemic bulbar stenosis.
1. Bilateral pudendal occlusion: revascularise before urethroplasty.
2. Bladder neck appearance does not predict continence after PFUI.

87 Surgery of the Scrotum and Seminal Vesicles

Reading time ≈ 32 min (slow pace)

1 · SURGICAL ANATOMY OF THE SCROTUM

- **Skin**: rugated (attached to dartos), thin, elastic. Its laxity gives **poor tamponade**, so haematomas keep enlarging: meticulous haemostasis is mandatory.
- **Layers, testis outwards**: tunica albuginea → tunica vaginalis → internal spermatic fascia → cremaster → external spermatic fascia → dartos → skin. Dartos keeps necrotising fasciitis off the testes.
- **Transdermal absorption** ~**40×** other skin; rich in 5α-reductase. MCQ

Territory	Artery	Parent
Scrotum, anterior	Superficial and deep external pudendal	Femoral
Scrotum, posterior	Posterior scrotal	Perineal (internal pudendal)
Testis, primary	Testicular	Aorta
Testis, secondary	Deferential	Superior vesical ← internal iliac
Testis, secondary	Cremasteric	Inferior epigastric ← external iliac

KEY POINT Three trunks (aorta, internal iliac, external iliac) anastomose in the cord (~10 cm above testis) and at the cauda. Hence the testicular artery can be ligated in Fowler–Stephens orchidopexy, and the superior epididymal artery sacrificed in vasectomy reversal.

- **Somatic nerves**: genital branch of genitofemoral (injury loses cremasteric reflex); ilioinguinal (anterior scrotum); posterior scrotal from pudendal.

2 · PERI-OPERATIVE PRINCIPLES AND ACCESS

- Hair removal with **razor on the day of surgery** (less scrotal trauma than clippers).
- **Antibiotics** (American Urological Association, AUA): single intravenous (IV) cephalosporin or clindamycin **only with risk factors** (e.g. age, smoking, steroids, immunodeficiency, colonised material).
- Local infiltration + spermatic cord block allows office surgery; pre-incisional 0.5% lidocaine (no adrenaline) reduces post-op pain.

Benign disease	Possible malignancy
Scrotal incision (median raphe = avascular plane, or transverse); open tunica directly	Inguinal incision; clamp cord with non-crushing clamp before handling testis (prevents seeding) → frozen section; spare testis if benign

- **Always close the dartos**: most haematomas come from inadequate dartos closure.

3 · SCROTAL WALL: EXCISION, SCROTECTOMY, RECONSTRUCTION

- Infected sebaceous cyst: drain, excise after resolution (high recurrence).
- **Fournier gangrene (FG)**: commonest indication for partial scrotectomy. Fluids + broad-spectrum antibiotics + **urgent debridement** (spreads **2–3 cm/h**); mortality ~**20%**. Re-debride every **24–48 h** until no necrosis. Anorectal extension may need colostomy.
- **Scrotal violation in non-seminomatous germ cell tumour (NSGCT)** → partial scrotectomy of the scar.
- If >**50–60%** skin lost: no immediate reconstruction; testes in medial thigh pockets, saline gauze or within vacuum-assisted closure (VAC).

ALGORITHM
Defect <50%, no tension → primary closure · <50% with tension → local scrotal advancement flap / secondary intention · >50% → VAC, then stretched scrotal advancement or delayed primary closure → no local tissue → fasciocutaneous thigh flap (superomedial/anterolateral) · meshed split-thickness skin graft (STSG) last.

KEY POINT Why grafts last: the scrotum works because it is lax; graft contracture defeats that, whereas residual scrotal skin stretches enormously and is the best donor.

4 · HYDROCELE

- Fluid between parietal and visceral tunica vaginalis. **Congenital**: communicating via patent processus; ~**80%** resolve by 18 months. **Acquired**: production–absorption imbalance; men >40 y.
- Treat only for discomfort, cosmesis, underlying malignancy; pyocele urgently. **Ultrasound with colour Doppler first**: excludes intratesticular tumour and decides scrotal versus inguinal route.

Technique	Steps	Best for
Lord plication	Deliver testis through small opening; radial plicating sutures; no dissection	Thin-walled, smaller; least haematoma, no drain
Jaboulay (Winkelman)	Excise part of parietal layer, evert remainder behind testis	Thin, floppy sacs
Bottleneck	Trim all but 2-cm cuff around testis/cord, tack edges	Excisional
Excision	Excise sac, oversew edges	Thick-walled, large, multiloculated, chronic; lowest recurrence

KEY POINT More tunica excised → lower recurrence but larger raw surface and haematoma risk; plication trades a little recurrence for a dry field. Recurrence after any successful repair is **0–6%**.

Aspiration and sclerotherapy (AS)

- For unfit or surgery-averse men. **Aspiration alone recurs** → always add sclerosant (doxycycline, tetracycline, alcohol, polidocanol).
- Steps: cord block → 16–19 G plastic angiocatheter, closed system → drain fully → check fluid for sperm (± culture/cytology) → sclerosant into collapsed sac, massage.
- Single-session success modest; **retreatment** brings resolution to **85–96%**. Poor in multiloculated or >**750 mL** sacs. Fewer complications than surgery, lower satisfaction.

5 · SPERMATOCELE AND EPIDIDYMAL SURGERY

- Spermatocele: operate only for symptoms. **Counsel on obstructive azoospermia**, especially corpus/cauda cysts.
- Steps: scrotal incision → open tunica vaginalis → dissect cyst to stalk → ligate (5-0/6-0); if not separable, unroof.
- **Epididymal tumour**: suspected malignancy → inguinal route with cord clamp and frozen section; biopsy-proven benign → scrotal microsurgical excision.
- **Epididymectomy**: Results **poor for chronic epididymitis** (>6 weeks) → last resort; better with a **palpable lesion**. Steps: ligate efferent tubules at caput → dissect corpus/cauda off testis → ligate stumps (limits sperm extravasation). Counsel that orchidectomy may follow.

6 · VARICOCELE REPAIR

- Present in ~**15%** of men. For orchialgia (dull ache on standing), **microsurgical repair** gives more complete relief.

Approach	Principle	Pros	Cons
†Retroperitoneal (Palomo) / laparoscopic	High ligation of internal spermatic vein	Few veins at this level	Misses cremasteric/external spermatic collaterals; artery and lymphatics often taken → hydrocele
†Inguinal (Ivanissevich)	Open canal, ligate cord veins	Access to external spermatic veins	Opens external oblique; higher pain
†Subinguinal microsurgical	Incision at external ring; microscope + micro-Doppler; ligate all veins, preserve artery, lymphatics, vas	**Lowest recurrence and hydrocele**; no fascial incision → least pain; gold standard	More veins and small arteries at this level; needs microsurgical skill
†Percutaneous embolisation	Venography + coils/sclerosant via femoral or jugular vein	Local anaesthesia, fast recovery; suits post-surgical recurrence	Cannulation failure; radiation; coil migration

7 · VASECTOMY

- **Counsel**: not immediately sterile (contraception until post-vasectomy semen analysis, PVSA); repeat vasectomy **<1%**; pregnancy ~**1 in 2000** (0.05%) even after azoospermic PVSA; chronic pain ~**1–2%**.
- **No routine antibiotics**. Lidocaine ± bupivacaine via 25–32 G needle; topical cream never sole anaesthetic.
- Two separate steps: **isolation** (determines morbidity) and **occlusion** (determines efficacy).

Feature	Conventional (CV)	Minimally invasive (MIV)	No-scalpel (NSV)
Skin opening	1.5–3 cm, median or bilateral	≤**10 mm**	Puncture <10 mm
Status	Difficult cases (prior surgery, thick skin)	**Recommended**: less pain, bleeding, infection	Original MIV (Li Shunqiang, China, 1974) MCQ

- **Three-finger grasp** brings vas under skin. Open at/above penoscrotal junction on the **straight vas**: convoluted vas is harder now and at reversal.
- **NSV steps**: ring clamp fixes vas under taut skin → sharp mosquito punctures skin and sheath, spreads → lifts vas out → strip sheath, freeing vasal artery → occlude → close or leave open. NSV defines isolation only, not occlusion.

AUA-recommended occlusion (failure ≤1%)	Note
Mucosal cautery (MC) + fascial interposition (FI)	FI = internal spermatic fascia over one end
MC alone (no clips/ligatures)	Cautery of lumen 0.5–1.5 cm
Open-ended: abdominal end MC + FI, testicular end open	Less congestive epididymal pain; sperm granuloma linked to better reversal

> **KEY POINT** Every recommended method rests on **mucosal cautery**; ligatures and clips alone are not core methods. Division of the vas is required; segment excision optional (**1 cm** adequate).

Post-vasectomy semen analysis

ALGORITHM

PVSA at **8–16 weeks** (after ≥10–20 ejaculations) → azoospermia or rare non-motile sperm (RNMS, ≤**100,000/mL**) → stop other contraception · motile sperm → repeat at 6 months → still motile = failure → repeat vasectomy.

- **Early failure**: technical (usually one vas not occluded), detected on PVSA. **Late failure**: recanalisation after successful PVSA (~1 in 2000).
- **Complications**: haematoma, infection, **sperm granuloma** (expected, not a complication; symptomatic up to 3 months → non-steroidal anti-inflammatory drugs (NSAIDs); persisting <5% → excise vasal ends/redo), post-vasectomy pain ~**2%** (cord block → denervation or reversal).

8 · ORCHIDECTOMY AND PROSTHESIS

- **Bilateral simple orchidectomy**: hormone-sensitive metastatic prostate cancer, avoiding medical androgen deprivation therapy (ADT); cheaper, less gynaecomastia; survival benefit unproven.
- **Unilateral**: severe trauma, non-salvageable torsion; chronic orchialgia only after all conservative measures fail.
- **Simple (scrotal)**: open all layers → divide cord in two bundles between clamps → suture-ligate stump.
- **Epididymis-sparing (subepididymal)**: testis dissected off epididymis under microscope → better contour and satisfaction. †Subcapsular orchidectomy (enucleate parenchyma, leave albuginea) has the same cosmetic aim.
- **For orchialgia use the inguinal route**: it removes more proximal ilioinguinal, iliohypogastric and genitofemoral nerve segments; persisting pain → resect genital branch of genitofemoral.
- **Prosthesis**: offer every man lacking a testis, before cancer orchidectomy. Silicone, saline-filled; pierce port ≤**5 times**, never post-op. Inguinal at radical orchidectomy/children; **high scrotal subdartos** pouch has least infection/erosion (implant away from suture line); trans-scrotal has higher extrusion.

9 · TESTICULAR TORSION AND FIXATION

- †**Acute torsion**: emergency scrotal exploration without delaying for imaging when clinical suspicion is high → detorse, warm, assess colour/bleeding → viable: fix; non-viable: orchidectomy → **always fix the contralateral testis**.
- **Intermittent torsion** (pain <2 h, spontaneous resolution) and postpubertal retractile testis → elective orchidopexy.

Suture fixation	Dartos pouch
Albuginea fixed to dartos (3-0 absorbable)	Subdartos tunnel to dependent scrotum; purse-string around cord at cremaster exit
Re-torsion possible years later even with non-absorbable; lower-pole sutures can infarct testis; transparenchymal sutures → inflammation, abscess, loss of spermatogonia (animals)	No transparenchymal suture; spermatogenesis normal (animals) → **preferred when fertility matters**

10 · CHRONIC SCROTAL PAIN

Pain >3 months interfering with life.

ALGORITHM
History, examination, duplex ultrasound → treat cause · none → exclude extrascrotal cause (cross-sectional imaging of abdomen, pelvis, spine, hip) → prostatitis/pelvic floor → treat · scrotal non-specific → support, NSAIDs, tricyclics, gabapentin ≥3 months → epididymal → epididymectomy · testis/post-vasectomy → spermatic cord block: >**50%** relief → microsurgical denervation (or reversal); <50% → pain clinic, inguinal orchidectomy last · post-hernia → ilioinguinal block >**90%** → neurolysis.

- **Microsurgical denervation of spermatic cord (MDSC)**: subinguinal 4-cm incision → divide ilioinguinal and genital branch of genitofemoral → microscope + micro-Doppler → **preserve testicular and cremasteric arteries, lymphatics (prevent hydrocele) and vas with vessels** → divide everything else (veins, cremaster, perivasal tissue).

- **Trifecta nerve complex**: cremaster fibres → perivasal tissue → posterior periarterial tissue/cord lipoma; the last must also be divided.

11 · COMPLICATIONS OF SCROTAL SURGERY

- **Haematoma** (commonest): drain if large, infected or enlarging. Ongoing bleeding after orchidectomy = slipped spermatic artery ligature → subinguinal incision, retrieve and re-ligate; falling haematocrit/expanding mass = **retroperitoneal bleed** → extend incision into retroperitoneum.
- **Infection**: antibiotics or open, irrigate, evacuate; leave open.
- **Iatrogenic obstruction** of epididymis/vas in hydrocele or spermatocele surgery → azoospermia.

12 · SEMINAL VESICLES: ANATOMY AND DISORDERS

- Seminal vesicles (SV): paired Wolffian (mesonephric) derivatives, bud at ~**12 weeks**; produce ~**60%** of seminal fluid. Join the ampulla to form the ejaculatory duct.
- Relations: bladder in front, rectum behind, **neurovascular bundles lateral to tips**, ureter crossing anterolateral to tips.
- **Magnetic resonance imaging (MRI) is most accurate**; transrectal ultrasound (TRUS) suggests obstruction if SV width >**1.5 cm** or ejaculatory duct >**2.3 mm** — but confirm by **SV aspiration showing sperm**.

Cysts

- Rare (~0.005%). Congenital (Wolffian anomaly) or acquired (ejaculatory duct obstruction from infection or stones).
- **Zinner syndrome**: SV cyst + **ipsilateral renal agenesis** + ejaculatory duct obstruction (often ectopic ureter into cyst) — † because ureteric bud and SV both arise from the mesonephric duct.
- Small/asymptomatic → observe or aspirate; **transurethral unroofing** (~1 cm^3 between bladder neck and verumontanum) is preferred for cysts, ejaculatory duct obstruction and stones; large symptomatic → laparoscopic/robotic excision.
- **Stones/haemospermia**: transurethral seminal vesiculoscopy (semirigid ureteroscope) with holmium laser. Re-obstruction common.

Tumours

Benign	Malignant
Cystadenoma, papillary adenoma, hydatid cyst, amyloid	Adenocarcinoma, mixed epithelial–stromal tumour, phyllodes, sarcoma, carcinoid, seminoma, yolk sac tumour

- Complex retrovesical mass; MRI → TRUS biopsy if inconclusive; know malignancy pre-op. †Exclude secondary invasion (prostate, bladder, rectum) first — far commoner than primary SV cancer.
- Adenocarcinoma presents late → radical excision + lymphadenectomy ± adjuvant therapy.

13 · SURGICAL APPROACHES TO THE SEMINAL VESICLES

Approach	Use	Main risk
Transvesical transtrigonal	Large benign cyst; ectopic ureter into cyst	Urinary leak, ureter
Perivesical (extravesical)	Children, large cyst ± nephroureterectomy	Ureter (superior), vas (superomedial)
Transperitoneal (laparoscopic/robotic)	Bilateral lesions; now dominant	Access injuries
Perineal	At perineal prostatectomy	Neurovascular bundle; incomplete excision

Approach	Use	Main risk
Transcoccygeal (prone jackknife)	Hostile abdomen/perineum	Rectum

- **Transvesical**: midline incision → anterior cystotomy → ureteric catheters → trigonal incision → traction on vas identifies SV → excise at vas junction → two-layer closure, drain, catheter.
- **Transperitoneal**: open peritoneum at rectovesical cul-de-sac → ampullae midline, SVs lateral. Opens neither organ, hence preferred.

Injury	Management
Rectum	No gross spillage → debride, two-layer repair; extensive/spillage → diverting colostomy
Ureter	Confirm with IV indigo carmine or retrograde study → ureteroneocystostomy ± psoas hitch/Boari flap, stent
Neurovascular bundle	Prevent by staying medial → erectile dysfunction

14 · MCQ ONE-LINERS

1. Commonest scrotal surgery complication: haematoma — prevented by dartos closure.
2. Dartos fascia limits Fournier gangrene spread to testes.
3. Injury to genital branch of genitofemoral: loss of cremasteric reflex.
4. Suspected testicular tumour: inguinal approach, cord clamp before handling testis.
5. Fournier gangrene spreads 2–3 cm/hour; re-debride every 24–48 h.
6. Lord plication: least haematoma, no drain, thin-walled hydrocele.
7. Jaboulay: partial excision and eversion of parietal tunica behind testis.
8. Sclerotherapy poor for multiloculated or >750 mL hydroceles.
9. Hydrocele recurrence after surgery: 0–6%, regardless of technique.
10. Spermatocele excision risk: obstructive azoospermia.
11. No-scalpel vasectomy: Li Shunqiang, China, 1974.
12. Recommended vas occlusion methods all use mucosal cautery.
13. Excised vas segment, if taken: 1 cm adequate.
14. PVSA at 8–16 weeks; RNMS ≤100,000 non-motile sperm/mL = success.
15. Pregnancy after successful vasectomy ~1 in 2000.
16. Sperm granuloma is not a complication.
17. Orchialgia: inguinal orchidectomy gives better pain outcome than scrotal.
18. Cord block >50% relief selects for microsurgical cord denervation.
19. Dartos pouch preferred over suture fixation for fertility.
20. Testicular prosthesis port: ≤5 punctures, never post-op.
21. Subinguinal microsurgical varicocelectomy: lowest recurrence and hydrocele †.
22. Seminal vesicles contribute ~60% of ejaculate volume.
23. Zinner syndrome: SV cyst + ipsilateral renal agenesis + ejaculatory duct obstruction.
24. TRUS obstruction: SV >1.5 cm, ejaculatory duct >2.3 mm; confirm by aspiration.

★ Putting it together: testis, penis and urethra decisions

Reading time ≈ 6 min (slow pace) · Integrates Ch 78–87 · use these as the skeleton of long-essay answers

A · TESTICULAR MASS: THE FIRST WEEK

ALGORITHM

Scrotal ultrasound → tumour markers (alpha-fetoprotein [AFP], human chorionic gonadotrophin [hCG], lactate dehydrogenase [LDH]) **before** orchidectomy → offer sperm banking → **radical inguinal orchidectomy** (never trans-scrotal) → repeat markers after orchidectomy (AFP half-life 5–7 days, hCG 1–1.5 days) → CT abdomen–pelvis and chest imaging → stage and risk group (→ *Ch 79, 80*).

B · GERM CELL TUMOUR MANAGEMENT BY STAGE

Stage	Seminoma	Non-seminoma (NSGCT)
Clinical stage I	Surveillance preferred (occult spread 10–15%); single-dose carboplatin as an alternative	Low risk (no lymphovascular invasion, not embryonal-predominant) → surveillance. High risk → surveillance, one cycle of bleomycin–etoposide–cisplatin (BEP), or nerve-sparing retroperitoneal lymph node dissection (RPLND)
Stage II	Nodes ≤3 cm: radiotherapy, chemotherapy or RPLND. Bulky >3 cm → chemotherapy	Normal markers, small nodes → primary RPLND possible; otherwise chemotherapy
Advanced (International Germ Cell Cancer Collaborative Group, IGCCCG)	Good risk: BEP ×3 or etoposide–cisplatin (EP) ×4. Intermediate/poor risk: BEP ×4 (etoposide–ifosfamide–cisplatin [VIP] ×4 if lungs are a concern). Treat fast in a high-volume centre; full-dose cisplatin	
Residual mass after chemotherapy	>3 cm → fluorodeoxyglucose positron emission tomography (PET) (≥6 weeks after chemotherapy) guides resection	Any mass >1 cm → post-chemotherapy RPLND (teratoma does not respond to chemotherapy)

C · PENILE CANCER: PRIMARY AND GROIN

ALGORITHM

Primary: penile intraepithelial neoplasia → topical 5-fluorouracil or imiquimod, laser, glans resurfacing. Invasive → organ-sparing excision, glansectomy, partial or total penectomy with perineal urethrostomy; margins under 1 cm are acceptable (→ *Ch 82*).

Groin, impalpable (cN0): low risk (pTis/pTa/pT1a) → surveillance; intermediate/high risk → dynamic sentinel node biopsy or modified inguinal node dissection; a positive node → radical dissection.

Groin, palpable (cN+): fine-needle aspiration → positive → bilateral radical inguinal dissection; fixed or bulky nodes (cN3) → neoadjuvant chemotherapy (paclitaxel–ifosfamide–cisplatin, TIP) first; pelvic dissection if ≥2 inguinal nodes or extranodal extension (→ *Ch 84*).

D · ANTERIOR URETHRAL STRICTURE: WHICH OPERATION?

Stricture	Treatment
Short (<1.5 cm) bulbar, first time	Dilation or direct-vision internal urethrotomy once; if it recurs → urethroplasty

Stricture	Treatment
Bulbar 1–2 cm (up to 3–5 cm proximally)	Excision and primary anastomosis
Longer bulbar	Buccal (oral) mucosa graft onlay ± augmented anastomosis
Penile	Onlay graft or flap; staged repair if the bed is hostile, lichen sclerosus or failed hypospadias
Pan-urethral, exhausted options, frail	Perineal urethrostomy
Pelvic fracture urethral injury	Suprapubic catheter, then delayed anastomotic repair via the progressive perineal approach

In acute retention with a stricture, place a suprapubic catheter rather than blind dilation (→ *Ch 86*).